NATIONAL ACADEMIES Sciences Engineering Medicine

NATIONAL ACADEMIES PRESS Washington, DC

This PDF is available at http://nap.nationalacademies.org/24659



NextGen for Airports, Volume 3: Resources for Airports (2016)

BUY THIS BOOK

FIND RELATED TITLES

DETAILS

94 pages | 8.5 x 11 | PAPERBACK

ISBN 978-0-309-44602-0 | DOI 10.17226/24659

CONTRIBUTORS

Robert Hemm, Virginia Stouffer, William Dunlay, Richard Marchi, Stephen Van Beek, Susan Sharp, and Mary Arzt; Airport Cooperative Research Program; Transportation Research Board; National Academies of Sciences, Engineering, and Medicine

SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2016. *NextGen for Airports, Volume 3: Resources for Airports*. Washington, DC: The National Academies Press. https://doi.org/10.17226/24659.

Visit the National Academies Press at nap.edu and login or register to get:

– Access to free PDF downloads of thousands of publications
– 10% off the price of print publications
– Email or social media notifications of new titles related to your interests
– Special offers and discounts

All downloadable National Academies titles are free to be used for personal and/or non-commercial academic use. Users may also freely post links to our titles on this website; non-commercial academic users are encouraged to link to the version on this website rather than distribute a downloaded PDF to ensure that all users are accessing the latest authoritative version of the work. All other uses require written permission. (Request Permission)

This PDF is protected by copyright and owned by the National Academy of Sciences; unless otherwise indicated, the National Academy of Sciences retains copyright to all materials in this PDF with all rights reserved.

ACRP REPORT 150

NextGEN for Airports

AIRPORT COOPERATIVE RESEARCH PROGRAM

Volume 3

Resources for Airports



TRB TRANSPORTATION RESEARCH BOARD

The National Academies of

SCIENCES • ENGINEERING • MEDICINE

Copyright National Academy of Sciences. All rights reserved.

ACRP OVERSIGHT COMMITTEE*

CHAIR

Kitty Freidheim
Freidheim Consulting

VICE CHAIR

Kelly Johnson
Northwest Arkansas Regional Airport Authority

MEMBERS

Gloria Bender
TransSolutions
Thella Bowens
San Diego International Airport
Benito De Leon
Federal Aviation Administration
Deborah Flint
Los Angeles World Airports
F. Paul Martinez
AvAOL, LLC
Scott McMahon
Morristown Municipal Airport
Frank Miller
San Antonio International Airport
Bob Montgomery
Southwest Airlines
Eric Potts
Freese and Nichols, Inc.
Megan S. Ryerson
University of Pennsylvania

EX OFFICIO MEMBERS

Sabrina Johnson
U.S. Environmental Protection Agency
Laura McKee
Airlines for America
Christopher Oswald
Airports Council International—North America
Neil J. Pedersen
Transportation Research Board
Gregory Principato
National Association of State Aviation Officials
Melissa Sabatine
American Association of Airport Executives
T.J. Schulz
Airport Consultants Council

SECRETARY

Christopher W. Jenks
Transportation Research Board

* Membership as of January 2016.

TRANSPORTATION RESEARCH BOARD 2016 EXECUTIVE COMMITTEE*

OFFICERS

CHAIR: **James M. Crites**, *Executive Vice President of Operations, Dallas–Fort Worth International Airport, TX*
VICE CHAIR: **Paul Trombino III**, *Director, Iowa Department of Transportation, Ames*
EXECUTIVE DIRECTOR: **Neil J. Pedersen**, *Transportation Research Board*

MEMBERS

Victoria A. Arroyo, *Executive Director, Georgetown Climate Center; Assistant Dean, Centers and Institutes; and Professor and Director, Environmental Law Program, Georgetown University Law Center, Washington, DC*
Scott E. Bennett, *Director, Arkansas State Highway and Transportation Department, Little Rock*
Jennifer Cohan, *Secretary, Delaware DOT, Dover*
Malcolm Dougherty, *Director, California Department of Transportation, Sacramento*
A. Stewart Fotheringham, *Professor, School of Geographical Sciences and Urban Planning, Arizona State University, Tempe*
John S. Halikowski, *Director, Arizona DOT, Phoenix*
Susan Hanson, *Distinguished University Professor Emerita, Graduate School of Geography, Clark University, Worcester, MA*
Steve Heminger, *Executive Director, Metropolitan Transportation Commission, Oakland, CA*
Chris T. Hendrickson, *Hamerschlag Professor of Engineering, Carnegie Mellon University, Pittsburgh, PA*
Jeffrey D. Holt, *Managing Director, Power, Energy, and Infrastructure Group, BMO Capital Markets Corporation, New York*
S. Jack Hu, *Vice President for Research and J. Reid and Polly Anderson Professor of Manufacturing, University of Michigan, Ann Arbor*
Roger B. Huff, *President, HGLC, LLC, Farmington Hills, MI*
Geraldine Knatz, *Professor, Sol Price School of Public Policy, Viterbi School of Engineering, University of Southern California, Los Angeles*
Ysela Llort, *Consultant, Miami, FL*
Melinda McGrath, *Executive Director, Mississippi DOT, Jackson*
James P. Redeker, *Commissioner, Connecticut DOT, Newington*
Mark L. Rosenberg, *Executive Director, The Task Force for Global Health, Inc., Decatur, GA*
Kumares C. Sinha, *Olson Distinguished Professor of Civil Engineering, Purdue University, West Lafayette, IN*
Daniel Sperling, *Professor of Civil Engineering and Environmental Science and Policy; Director, Institute of Transportation Studies, University of California, Davis*
Kirk T. Steudle, *Director, Michigan DOT, Lansing*
Gary C. Thomas, *President and Executive Director, Dallas Area Rapid Transit, Dallas, TX*
Pat Thomas, *Senior Vice President of State Government Affairs, United Parcel Service, Washington, DC*
Katherine F. Turnbull, *Executive Associate Director and Research Scientist, Texas A&M Transportation Institute, College Station*
Dean Wise, *Vice President of Network Strategy, Burlington Northern Santa Fe Railway, Fort Worth, TX*

EX OFFICIO MEMBERS

Thomas P. Bostick (Lieutenant General, U.S. Army), *Chief of Engineers and Commanding General, U.S. Army Corps of Engineers, Washington, DC*
James C. Card (Vice Admiral, U.S. Coast Guard, retired), *Maritime Consultant, The Woodlands, Texas, and Chair, TRB Marine Board*
T. F. Scott Darling III, *Acting Administrator and Chief Counsel, Federal Motor Carrier Safety Administration, U.S. DOT*
Marie Therese Dominguez, *Administrator, Pipeline and Hazardous Materials Safety Administration, U.S. DOT*
Sarah Feinberg, *Administrator, Federal Railroad Administration, U.S. DOT*
Carolyn Flowers, *Acting Administrator, Federal Transit Administration, U.S. DOT*
LeRoy Gishi, *Chief, Division of Transportation, Bureau of Indian Affairs, U.S. Department of the Interior, Washington, DC*
John T. Gray II, *Senior Vice President, Policy and Economics, Association of American Railroads, Washington, DC*
Michael P. Huerta, *Administrator, Federal Aviation Administration, U.S. DOT*
Paul N. Jaenichen, Sr., *Administrator, Maritime Administration, U.S. DOT*
Bevan B. Kirley, *Research Associate, University of North Carolina Highway Safety Research Center, Chapel Hill, and Chair, TRB Young Members Council*
Michael P. Melaniphy, *President and CEO, American Public Transportation Association, Washington, DC*
Gregory G. Nadeau, *Administrator, Federal Highway Administration, U.S. DOT*
Wayne Nastri, *Acting Executive Officer, South Coast Air Quality Management District, Diamond Bar, CA*
Mark R. Rosekind, *Administrator, National Highway Traffic Safety Administration, U.S. DOT*
Craig A. Rutland, *U.S. Air Force Pavement Engineer, U.S. Air Force Civil Engineer Center, Tyndall Air Force Base, FL*
Reuben Sarkar, *Deputy Assistant Secretary for Transportation, U.S. Department of Energy*
Gregory D. Winfree, *Assistant Secretary for Research and Technology, Office of the Secretary, U.S. DOT*
Frederick G. (Bud) Wright, *Executive Director, American Association of State Highway and Transportation Officials, Washington, DC*
Paul F. Zukunft (Admiral, U.S. Coast Guard), *Commandant, U.S. Coast Guard, U.S. Department of Homeland Security*

* Membership as of April 2016.

Copyright National Academy of Sciences. All rights reserved.

AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP REPORT 150

NextGEN for Airports Volume 3

Resources for Airports

Robert Hemm
Virginia Stouffer
LMI
Tysons, VA

William Dunlay
LEIGH-FISHER
Burlingame, California

Richard Marchi
RF MARCHI AVIATION
Washington, D.C.

Stephen Van Beek
STEER DAVIES GLEAVE
Washington D.C.

Susan Sharp
Mary Arzt
SHARP AND COMPANY
Rockville, MD

Subscriber Categories
Aviation, Environment, Planning and Forecasting

Research sponsored by the Federal Aviation Administration

TRANSPORTATION RESEARCH BOARD

WASHINGTON, D.C.
2016
www.TRB.org

Copyright National Academy of Sciences. All rights reserved.

AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in TRB Special Report 272: Airport Research Needs: Cooperative Solutions in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

ACRP benefits from the cooperation and participation of airport professionals, air carriers, shippers, state and local government officials, equipment and service suppliers, other airport users, and research organizations. Each of these participants has different interests and responsibilities, and each is an integral part of this cooperative research effort.

Research problem statements for ACRP are solicited periodically but may be submitted to TRB by anyone at any time. It is the responsibility of the AOC to formulate the research program by identifying the highest priority projects and defining funding levels and expected products.

Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

ACRP REPORT 150, Volume 3

Project 01-27
ISSN 1935-9802
ISBN 978-0-309-44602-0
Library of Congress Control Number 2016941099

© 2016 National Academy of Sciences. All rights reserved.

COPYRIGHT INFORMATION

Authors herein are responsible for the authenticity of their materials and for obtaining written permissions from publishers or persons who own the copyright to any previously published or copyrighted material used herein.

Cooperative Research Programs (CRP) grants permission to reproduce material in this publication for classroom and not-for-profit purposes. Permission is given with the understanding that none of the material will be used to imply TRB, AASHTO, FAA, FHWA, FMCSA, FRA, FTA, Office of the Assistant Secretary for Research and Technology, PHMSA, or TDC endorsement of a particular product, method, or practice. It is expected that those reproducing the material in this document for educational and not-for-profit uses will give appropriate acknowledgment of the source of any reprinted or reproduced material. For other uses of the material, request permission from CRP.

NOTICE

The report was reviewed by the technical panel and accepted for publication according to procedures established and overseen by the Transportation Research Board and approved by the National Academies of Sciences, Engineering, and Medicine.

The opinions and conclusions expressed or implied in this report are those of the researchers who performed the research and are not necessarily those of the Transportation Research Board; the National Academies of Sciences, Engineering, and Medicine; or the program sponsors.

The Transportation Research Board; the National Academies of Sciences, Engineering, and Medicine; and the sponsors of the Airport Cooperative Research Program do not endorse products or manufacturers. Trade or manufacturers' names appear herein solely because they are considered essential to the object of the report.

Published reports of the

AIRPORT COOPERATIVE RESEARCH PROGRAM

are available from

Transportation Research Board
Business Office
500 Fifth Street, NW
Washington, DC 20001

and can be ordered through the Internet by going to

http://www.national-academies.org

and then searching for TRB

Printed in the United States of America

Copyright National Academy of Sciences. All rights reserved.

The National Academies of
SCIENCES • ENGINEERING • MEDICINE

The **National Academy of Sciences** was established in 1863 by an Act of Congress, signed by President Lincoln, as a private, non-governmental institution to advise the nation on issues related to science and technology. Members are elected by their peers for outstanding contributions to research. Dr. Marcia McNutt is president.

The **National Academy of Engineering** was established in 1964 under the charter of the National Academy of Sciences to bring the practices of engineering to advising the nation. Members are elected by their peers for extraordinary contributions to engineering. Dr. C. D. Mote, Jr., is president.

The **National Academy of Medicine** (formerly the Institute of Medicine) was established in 1970 under the charter of the National Academy of Sciences to advise the nation on medical and health issues. Members are elected by their peers for distinguished contributions to medicine and health. Dr. Victor J. Dzau is president.

The three Academies work together as the **National Academies of Sciences, Engineering, and Medicine** to provide independent, objective analysis and advice to the nation and conduct other activities to solve complex problems and inform public policy decisions. The Academies also encourage education and research, recognize outstanding contributions to knowledge, and increase public understanding in matters of science, engineering, and medicine.

Learn more about the National Academies of Sciences, Engineering, and Medicine at www.national-academies.org.

The **Transportation Research Board** is one of seven major programs of the National Academies of Sciences, Engineering, and Medicine. The mission of the Transportation Research Board is to increase the benefits that transportation contributes to society by providing leadership in transportation innovation and progress through research and information exchange, conducted within a setting that is objective, interdisciplinary, and multimodal. The Board's varied committees, task forces, and panels annually engage about 7,000 engineers, scientists, and other transportation researchers and practitioners from the public and private sectors and academia, all of whom contribute their expertise in the public interest. The program is supported by state transportation departments, federal agencies including the component administrations of the U.S. Department of Transportation, and other organizations and individuals interested in the development of transportation.

Learn more about the Transportation Research Board at www.TRB.org.

Copyright National Academy of Sciences. All rights reserved.

COOPERATIVE RESEARCH PROGRAMS

CRP STAFF FOR ACRP REPORT 150, Volume 3

Christopher J. Hedges, *Director, Cooperative Research Programs*
Michael R. Salamone, *ACRP Manager*
Jeffrey Oser, *Program Coordinator*
Eileen P. Delaney, *Director of Publications*
Scott E. Hitchcock, *Editor*

ACRP PROJECT 01-27 PANEL
Field of Administration

Frederick R. Busch, *Denver International Airport (Retired), Denver, CO* (Chair)
Brian Gonzalez, *American Airlines, Miami, FL*
Pamela S. Keidel-Adams, *Kimley-Horn and Associates, Phoenix, AZ*
Chad E. Leqve, *Minneapolis-St. Paul Metropolitan Airports Commission, Minneapolis, MN*
Erik R. Treudt, *Tallahassee International Airport, Tallahassee, FL*
James A. Wilding, *Glenwood, MD*
Tony Diana, *FAA Liaison*
Beth White, *FAA Liaison*
Christopher J. Oswald, *Airports Council International - North America Liaison*
Christine Gerencher, *TRB Liaison*

Copyright National Academy of Sciences. All rights reserved.

Foreword

By **Michael R. Salamone**
Staff Officer
Transportation Research Board

The Next Generation Air Transportation System (NextGen) refers to the federal programs (predominately airspace, air traffic, or avionics related) that are designed to modernize the National Airspace System (NAS). ACRP's NextGen initiative aims to inform airport operators about some of these programs and how the enabling practices, data, and technologies resulting from them will affect airports and change how they operate.

ACRP Report 150: NextGen for Airports, Volume 3: Resources for Airports presents a comprehensive overview of the many elements of NextGen, in terms and context that are relevant, familiar, and understandable to airport operators. The audience for this research report includes airport staff with a working knowledge of airports and the report is formulated to convey relevant information and technical guidance to airports. It is also intended to provide planning guidance for airport directors, department heads, board members, and other senior policy interests within the airport organization.

NextGen is an extensive and critical component of the future of the NAS that currently encompasses more than 100 separate program, technology, and procedure changes, some directly affecting airports, others indirectly, and others not at all. Because of its technical nature, however, and the fact that much of the published material about NextGen is geared to industry experts and not to airport leadership and airport stakeholders, little is known about NextGen and how it would change airport operations outside the FAA. This guide and associated primer broadly address potential airport impacts and include background information so that airport personnel can discern which NextGen programs will impact them and how.

Under ACRP Project 01-27 the research team, led by Logistics Management Institute, Inc., developed this resource guide, which provides a comprehensive list of NextGen technologies and initiatives categorized and described for airport practitioners and presents existing FAA plans that could potentially affect airports of all sizes and roles, the larger aviation industry, and the public. This guide presents a likely timeline for implementation and highlights the FAA's planned rollout of near- and mid-term elements as well as its long-range vision. Appendices include a glossary of terms and a Public Information Toolkit for communicating about the NextGen initiative to the broad spectrum of external stakeholders. This Public Information Toolkit is also available on the ACRP NextGen for Airports series website at https://crp.trb.org/acrp0127.

Copyright National Academy of Sciences. All rights reserved.

Copyright National Academy of Sciences. All rights reserved.

Contents

1 Chapter 1 **Introduction**

1 What Is NextGen?
2 Online Information Sources
4 What NextGen Does
4 Flight Efficiency
5 Fewer Flight Delays with Equivalent or Improved Safety
5 Environmental Emissions Impact
5 Electronic Data Benefits and Requirements
6 Fuel-Saving Flight Trajectories and Noise Impacts
6 Metroplex and Single Airport Access
7 Deferred Capital Improvements
7 Resource Guide Organization

8 Chapter 2 **NextGen Architecture**

9 Major NextGen Programs
10 PBN
11 Navigation History
16 Surface Operations and Data Sharing
17 MRO
18 Data Comm
19 Aircraft Equipment Mandate
20 Geographic Information System (GIS) Mandate and Airport Improvement Program (AIP)

22 Chapter 3 **NextGen Programs and Portfolios**

53 Chapter 4 **NextGen Airport by Airport**

53 General Considerations
54 Airports Near a Major Hub with Overlapping Approaches
55 Airports with One or No ILS
55 Airports with Multiple ILS
55 Airports with Predominantly GA Traffic
56 Airports Without Radar Surveillance
56 Airports Planning Improvements

Copyright National Academy of Sciences. All rights reserved.

59 Chapter 5 **Working with the FAA**

61 Where NextGen EAs Originate in the FAA
61 NEPA (National Environmental Policy Act) Process Impacts
63 Summary

65 Appendix A Acronyms and Abbreviations

69 Appendix B Glossary

74 Appendix C Public Information Toolkit

Copyright National Academy of Sciences. All rights reserved.

1 Introduction

What Is NextGen?

The Next Generation Air Transportation System (NextGen) is a set of technology modernization programs and procedure changes being developed and implemented over the next 15 years with the goal of improving the operation of the National Airspace System (NAS). NextGen began in December 2003 with the Vision 100 Century of Aviation Reauthorization Act.[1] Among its first goals were to

> . . . take advantage of data from emerging ground-based and space-based communications, navigation, and surveillance technologies;
>
> . . . integrate data streams from multiple agencies and sources to enable situational awareness and seamless global operations for all appropriate users of the system. . .[2]

The FAA has primary responsibility for NextGen development and implementation, through a NextGen headquarters (FAA/ANG) office separate from both Air Traffic Operations (ATO) and the FAA Airports headquarters office (FAA/ARP). The FAA publishes online information covering both the NextGen and Airport information that is referenced in the following sections.

Although the FAA is leading the NextGen public investment effort, NextGen success depends on many aviation users and service providers. NextGen programs affect various stakeholders in different ways. For example, in Figure 1-1, the FAA investments in NextGen performance-based navigation (PBN) procedures must be coupled with aircraft operator equipage in satellite-based navigation equipment, and accurate airport geospatial data to enable implementation. Importantly, this has an impact on airport communities through noise, and airport operators need to seek out plans of such changes, via the FAA websites, in order to provide the best outcome for the airport.

As examples of practical impacts, air traffic controllers are getting new computers and displays for air traffic control (ATC) that can display global positioning system (GPS)–based reports transmitted by aircraft using automatic dependent surveillance-broadcast (ADS-B) transceivers; aircraft owners can equip their aircraft with avionics to display electronic airport information and maps, approach and departure procedures, weather maps, and air traffic maps; and all stakeholders, including airports, can have access to real-time system status and weather information.

Infrastructure changes, combined with programs, procedures, and technologies, could potentially enable individual airport improvements, such as increased capacity multiple runway operations (MRO) and optimization of airspace and procedures for multi-airport Metroplexes.[3]

Airport personnel should be aware of these coming changes, which may be both beneficial and challenging, in order to participate in the public conversation, take advantage of technologies that will

[1] FAA, *A Brief History of the FAA,* www.faa.gov/about/history/brief_history.
[2] Vision 100—Century of Aviation Reauthorization Act, Public Law 108–176, December 12, 2003.
[3] A metroplex is a system of airports in proximity, along with their shared airspace, serving one or more major cities.

Copyright National Academy of Sciences. All rights reserved.

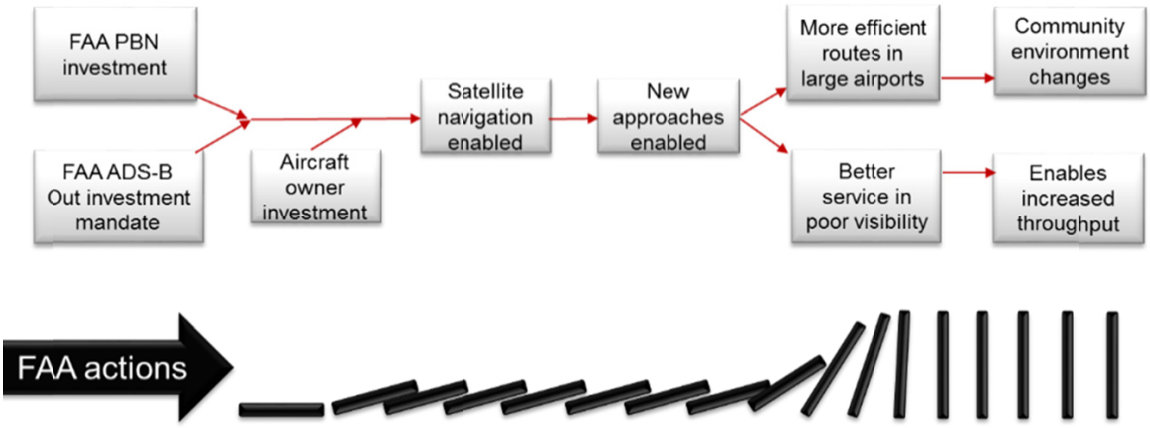


Figure 1-1. *The FAA makes NextGen investments and creates mandates, which interact in complex ways and have a domino-like impact on airports.*

benefit the airport, make sure the airport meets current FAA NextGen technology requirements or standards, and notify their communities of the changes.

This guidebook explains what NextGen is, what it does, the changes it brings, and the advantages it gives aviation users and providers. It also identifies, in layman's terms, ways that NextGen could affect airports. Moreover, because NextGen is a complex and ever-changing subject, this guide highlights major NextGen programs focusing on near-term FAA programs, procedures, and technologies, while describing long-term planning (beyond 5 years) in more general terms.

This guidebook, which is intended to present a high-level introduction to NextGen, is one of an ACRP series of five ACRP NextGen reports; the other four are listed below:

1. *ACRP Report 150: NextGen for Airports, Volume 1: Understanding the Airport's Role in Performance-Based Navigation: Resource Guide*
2. *ACRP Report 150: NextGen for Airports, Volume 2: Engaging Airport Stakeholders: Guidebook*
3. *ACRP Report 150: NextGen for Airports, Volume 4: Leveraging NextGen Spatial Data for Airports: Guidebook*
4. *ACRP Report 150: NextGen for Airports, Volume 5: Airport Planning and Development*

Online Information Sources

FAA: The FAA maintains an extensive array of online NextGen information that can be easily accessed using a web browser. For example, the query "FAA NextGen" brings up links to both a NextGen overview site, NextGen_Overview (https://www.faa.gov/nextgen/), and a NextGen Resources site, NextGen_Resources (https://www.faa.gov/nextgen/library/). The latter contains useful links to information on the NextGen infrastructure technologies, the NextGen Implementation Plan, and other NextGen related documents. The site https://www.faa.gov/nextgen/qanda/airports/ features airport-related questions and answers, and a further link to a "NextGen Works for Airports Brochure." It also

Copyright National Academy of Sciences. All rights reserved.

features "NextGen Priorities," NextGen_Priorities (https://www.faa.gov/nextgen/snapshots/priorities/), which includes status information on NextGen progress.

Radio Technical Commission for Aeronautics (RTCA): The FAA has established a joint government/industry Federal NextGen Advisory Committee (NAC) managed by RTCA, Inc. to foster industry collaboration on NextGen priorities and implementation. The link, NAC (http://www.rtca.org/nac) leads to the NAC site with links to NAC activities, membership, documents, and other information. The site also has links to other RTCA committees and RTCA documents.

NASA: NASA has been heavily involved for several years with developing aviation technologies related to NextGen. A "NASA NextGen" internet search leads to sites at NASA headquarters and NASA Ames Research Center. NASA research is focused on mid- and long-term technologies leading to gate-to-gate trajectory-based operations (TBO).[4] The NASA technical reports server at NASA_Tech_Reports (http://ntrs.nasa.gov/search.jsp) can be used to search and download NASA reports.

Conference Papers: Several organizations hold conferences and publish papers related to NextGen, including the Transportation Research Board (TRB). Of particular note are 11 USA/Europe Air Traffic Management Research and Development Seminars, ATM_Seminars (http://www.atmseminar.org/), that include presentations and papers from researchers from both the U.S. and abroad, many describing NextGen technologies in a research and measurement view. The papers and presentations from past seminars are all available online.

Regulatory Material: Aviation, including airports, is regulated by federal law primarily through Title 14 of the Code of Federal Regulations (14 CFR). Information available online includes the federal law, FAA orders, FAA advisory circulars, and other FAA guidance documents. Table 1-1 contains links to websites with links to airport-related regulatory information. NextGen programs have generated, and continue to generate, changes to the body of regulatory code describing the conduct of aviation.

Table 1-1. Regulatory websites.

TOPIC	WEBSITE	AIRPORT RELATED
Federal law	Federal_Code (https://www.gpo.gov/fdsys/granule/USCODE-2011-title49/USCODE-2011-title49-subtitleVII-partA-subparti-chap401-sec40101)	Title 49, Transportation, Part B - Airport Development and Noise, Part D – Public Airports
Federal regulations	Federal_Regs (http://www.ecfr.gov/cgi-bin/text-idx?SID=adcd42bf1eae8417436c765b02aae2f8&mc=true&tpl=/ecfrbrowse/Title14/14tab_02.tpl)	Title 14, Aeronautics and Space, Chapter 1, FAA, Subchapter I, Airports
FAA orders	FAA_Orders (https://www.faa.gov/regulations_policies/orders_notices/)	Link to Airport-related FAA Orders [Searching performance-based navigation generates links to PBN orders and notices produced by the FAA Aviation Safety (AVS) Flight Standards (AFS) and ATO Mission Support (AJV)]

[4]Time is sometimes substituted for trajectory signifying the shift from physical to time-based separation management.

Copyright National Academy of Sciences. All rights reserved.

Table 1-1. Continued

TOPIC	WEBSITE	AIRPORT RELATED
FAA advisory circulars	FAA_Advisory_Circulars (https://www.faa.gov/regulations_policies/advisory_circulars/)	Links to 150-series airport circulars

What NextGen Does

NextGen affects the NAS by modernizing the computers, automation, and electronics for communications, navigation, and surveillance with the goal of enabling new functions, setting a path for new technologies, enabling more efficient routes, and making other changes. For example, before NextGen, the NAS used only radar positioning for aircraft because air traffic computers could only accept radar inputs. New automation allows both radar and GPS position information.

NextGen is a very large program for the FAA, which has taken and will continue to take a major share of its capital investment plan budget through 2025. Two-thirds of the FAA's annual budget of $15 billion is spent keeping equipment operating and employing 15,000 air traffic controllers. The remaining funds are spent almost equally on new acquisitions such as NextGen and on-airport capital improvement projects. The NextGen computer replacement at 22 en route centers alone costs more than $2 billion.[5] Thus, the frequent news coverage of NextGen is partly because it occupies the attention of FAA headquarters. However, NextGen is projected to save much more than $2 billion for various users. The projected benefits and effects of NextGen fall into several categories:

Flight efficiency improvements,

Fewer flight delays in bad weather with equivalent or improved safety,

Environmental emissions benefits,

Electronic data benefits and requirements,

Fuel-saving flight trajectories and noise impacts,

Metroplex airport access, and

Deferred capital improvements.

Flight Efficiency

One reason for the attention paid to NextGen is that it is expected to increase flying efficiency and save fuel. In 2014, at 27 percent, fuel was commercial operators' single largest cost.[6] In that year, U.S. airlines spent nearly $30 billion on fuel.[7] Since 2014 fuel prices have dropped, but fuel remains a significant and demonstrably unstable cost item. Historically, the general aviation (GA) community pays nearly twice the price for a gallon of fuel than that paid by the commercial airlines. When fuel prices rise, requests for direct routing by GA pilots rise by a third, according to a National Business Aircraft Association study. Facilitating more direct routings and fuel-efficient airport arrivals and departures is a goal of new routes planned under NextGen.

[5]U.S. Department of Transportation, "Budget Estimates Fiscal Year 2013, Federal Aviation Administration," Committee Reports of the 113th Congress (2013–14), House Report 113-136, Departments of Transportation, and Housing and Urban Development, and Related Agencies Appropriations Bill.

[6]Airlines for America, U.S. Passenger Airline Cost Index (PACI), 2014.

[7]Bureau of Transportation Statistics, *Airline Fuel Cost and Consumption (U.S. Carriers—Scheduled)*, www.transtats.bts.gov/fuel.asp.

Copyright National Academy of Sciences. All rights reserved.

Fewer Flight Delays with Equivalent or Improved Safety

From 2008 to 2013 FAA data indicated that weather caused 69% of delays greater than 15 minutes.[8] In terminal areas, this often-quoted statistic means that ATC did not have an approach to use under the prevailing weather conditions that was able to maintain airport throughput, resulting in increased delay. This is particularly the case when independent operations to parallel runways that occur in visual conditions revert to single runway operations in instrument conditions. NextGen MRO technologies can reduce the current loss of capacity that occurs in bad weather and low-visibility conditions by enabling use of closely spaced parallel runways in marginal and instrument meteorological conditions. For single runways, GPS-based navigation approaches may provide an affordable way to improve airport all-weather operations and support improved airline schedule reliability for airports not having instrument landing systems (ILS) due to cost or challenging terrain, because the GPS-enabled approaches can provide both straight-in and curved precision approaches to new runways without the requirement for localizer (lateral) and glide slope (elevation) equipment.[9] The use of ADS-B Out technology will provide air traffic controllers with radar-like surveillance capability for certain airports currently without radar coverage. As part of NextGen, the FAA has mandated ADS-B Out equipage for all aircraft operating in certain controlled airspace, including all major airports, but ADS-B Out equipage is not universal and will likely be less prevalent at smaller GA airports. In lieu of ADS-B, wide-area multilateration (WAM) technology that interrogates current non-ADS-B transponders to provide radar-like surveillance has been deployed at a few smaller airports where mountainous terrain prevents radar surveillance.

Environmental Emissions Impact

More efficient en route and terminal flight paths reduce fuel burn and carbon dioxide emissions per flight. In addition, the growing use of bio-based jet fuels reduces aviation's net carbon dioxide contribution to the atmosphere per pound of fuel burned. Some new programs reduce ground taxiing and queuing time benefitting local air quality, again through reductions in fuel burn.

Electronic Data Benefits and Requirements

Electronic maps with accurate data showing the airfield layout, obstructions, flight tracks, and other data from the FAA, airports, local agencies, and vendors are converging to give airport operators, pilots, FAA procedure designers, and air traffic controllers greater situational awareness through an accurate common operating picture.[10]

For improved surface operations in low-visibility conditions, airports that do not already have an electronic map (indicating all physical features, including obstructions and markings) may have to create one as part of their airport master plan. Airports are responsible for submitting and updating an airport layout plan to reflect the configuration of the airfield, as well as notices to airmen (NOTAMs) when operational changes occur. When new flight procedures are designed or runway configurations changed, airports must submit specific safety-critical data. Increasingly, these data are submitted in an electronic form via the FAA's Airports Geographic (a.k.a., Geospatial) Information System (GIS) Program. Some airports may choose to place or require transponder equipment on vehicles to improve surveillance, tracking, and safety on the movement area.

[8]FAA FAQ: Weather Delay: https://www.faa.gov/nextgen/programs/.

[9]Precision approaches include both lateral and vertical guidance, while non-precision approaches only provide lateral guidance.

[10]For in-depth information on the many implications of electronic geographic mapping, see ACRP Report 150, *NextGen for Airports, Volume 4: Leveraging NextGen Spatial Data for Airports: Guidebook*. See also the FAA GIS website, http://www.faa.gov/airports/planning_capacity/airports_gis_electronic_alp/.

Copyright National Academy of Sciences. All rights reserved.

Fuel-Saving Flight Trajectories and Noise Impacts

Airports need to be prepared to work with the FAA and notify the community about changes in air routes that can change community aircraft noise. The process the FAA follows in changing NextGen airspace trajectories has differed from the environmental and public notification processes airports follow when undertaking projects such as new runways or airfield improvements. The FAA is authorized by public law to use a categorical exclusion (CatEx) environmental process for certain NextGen projects. The CatEx process assumes that environmental impacts are minimal and does not require formal public notification barring "extraordinary circumstances."[11] With PBN procedures, total noise exposure and emissions can be reduced, but air routes shift, noise impacts change, and communities have complained. A January 24, 2015, Associated Press article regarding new procedures, that redirected departures over the historic section of downtown Phoenix, reported the FAA position that the new procedures that have caused complaints "cannot be reverted back to the original routes," and that city council members were recommending that the city should consider filing a lawsuit and a historic preservation complaint.[12] On March 29, 2016, the FAA issued FAA Order 1050.1, Guidance Memo, "Guidance for Implementation of the Categorical Exclusion in Section 213(c)(2) of the FAA Modernization and Reform Act of 2012," which states that public notification provisions apply to this CatEx.[13] The memo states that, "Collaboration with airport operators and public notification should include provision for appropriate community outreach that not only informs the affected public of the FAA's proposal, but also allows the public to provide feedback on community concerns." *ACRP Report 150: NextGen for Airports, Volume 2: Engaging Airport Stakeholders: Guidebook* (ACRP Project 01-28), is a reference for this outreach, as are the outreach materials mentioned in this guidebook.[14]

Metroplex and Single Airport Access

NextGen-enabled trajectory changes are designed to create more efficient routes to reduce delay and fuel consumption for aircraft arriving and departing from both the major and non-major airports in a multi-airport Metroplex. Improved navigation accuracy based on route area navigation/required navigation performance (RNAV/RNP) flight paths could increase the number of routes into and out of the terminal airspace by allowing separate arrival and departure fixes for Metroplex airports that currently share a common fix, which could potentially increase both arrival and departure capacity at the airports. This capability has been applied to operations at Dallas-Ft. Worth and Dallas Love airports, and Chicago O'Hare and Chicago Midway airports.

RNAV (GPS) approaches can be developed for runways with or without existing ILS. As of May 26, 2016, there are 3,678 wide-area augmentation system (WAAS) localizer performance with vertical guidance (LPV) approach procedures serving 1,790 airports. 1,041 of these airports are non-ILS airports.[15] These approaches provide back-up for ILS outages and new capability for non-ILS runways. Airport operators should understand, though, that air carrier aircraft are not normally equipped with WAAS receivers that permit LPV approach procedures. In contrast, high-end GA and business aircraft operators are frequent users of this technology.

---

[11]Council on Environmental Quality, *A Citizen's Guide to the NEPA*, December 2007, pp 8-10.

[12]"FAA says Phoenix Sky Harbor flight paths cannot be reverted," Associated Press, ktar.com/22/1801635/FAA-says-Phoenix-Sky-Harbor-flight-paths-cannot-be-reverted.

[13]FAA Order 1050.1 Guidance Memo, *Guidance for Implementation of the Categorical Exclusion in Section 213(c)(2) of the FAA Modernization and Reform Act of 2012*, March 29, 2016.

[14]See also *ACRP Report 15: Aircraft Noise: A Toolkit for Managing Community Expectations*, Transportation Research Board of the National Academies of Sciences, Engineering, and Medicine, 2009; *ACRP Report 85: Developing and Maintaining Support for Your Airport Capacity Project*, Transportation Research Board of the National Academies of Sciences, Engineering, and Medicine, 2013; *ACRP Synthesis 17: Approaches to Integrating Airport Development and Federal Environmental Review Processes*, Transportation Research Board of the National Academies of Sciences, Engineering, and Medicine, 2009.

[15]FAA Website, Satellite Navigation – GPS/WAAS Approaches, https://www.faa.gov/about/office_org/headquarters_offices/ato/service_units/techops/navservices/gnss/approaches/.

Copyright National Academy of Sciences. All rights reserved.

Deferred Capital Improvements

Certain NextGen operational improvements expected to increase airport capacity, such as MRO and reduced aircraft separations, could defer or eliminate the need for capital improvements such as new runways and taxiways. Moreover, a particular capital improvement, such as a new runway, would be made much less expensive if it could be moved closer to adjacent existing runways, thereby potentially reducing the need for expensive demolition of facilities or property acquisition.

Resource Guide Organization

NextGen currently encompasses more than 100 separate program, technology, and procedure changes, some directly affecting airports, others indirectly, and others not at all. This guide addresses potential airport impacts and includes background information so that airport personnel can discern which NextGen programs will impact them and how. The remainder of this guide is organized as follows:

Chapter 2 describes the NextGen development and implementation "architecture," dividing the programs into four major groups. It explains NextGen changes in the context of the history of navigation.

Chapter 3 contains tabular fact sheets about major NextGen programs, portfolios, and technologies, with an analysis of impacts on airports. The tables identify the what, why, how, who, where, and when of each major NextGen program. They also identify what airports can do to take advantage of the program.

Chapter 4 highlights how NextGen programs will affect some airports more than others, emphasizing how they impact non-hub and GA airports.

Chapter 5 lists points of contact for airports to find more information about NextGen and when it may impact each airport. It also describes how environmental assessments (EAs) for NextGen programs differ from environmental review processes followed by airports.

Appendices A and B provide abbreviations used in the guide and a glossary of NextGen terms.

Appendix C is the Public Information Toolkit.

Copyright National Academy of Sciences. All rights reserved.

2 NextGen Architecture

NextGen encompasses both communication, surveillance, and navigation technologies (CNS) and a wide range of other related initiatives. Defining and tracking individual NextGen programs can be difficult, in part because NextGen is so large and many of the technologies are unfamiliar to airports. For instance, one program with a noticeable airport impact is the wake vortex recategorization (Wake RECAT) program, which reduces the required wake turbulence separations between certain aircraft pairs. Wake RECAT has been the subject of ongoing research by the FAA, International Civil Aviation Organization (ICAO), and commercial aircraft operators for several years. As a result of this collaborative work, the FAA has implemented a change in the required wake turbulence separations between landing and departing aircraft at certain participating airports, and has made this change a part of its NextGen program. The lack of an obvious technical relation between programs makes defining "NextGen" an organizational rather than logical grouping.

The content of NextGen is not static. From time to time, new programs are added to NextGen, as when a NASA air traffic management program that shows the promise of benefitting users demonstrates an adequate level of technical maturity is handed off to the FAA, and becomes a part of the FAA NextGen portfolio. The FAA also discontinues NextGen programs in favor of newer or better alternatives, such as cancellation of precision runway monitor (PRM) radar installations at airports in favor of parallel runway separation reductions under MRO without high update radar but with advanced controller monitoring displays.[16] As NextGen technologies are implemented the FAA will discontinue redundant installations to manage its operational maintenance burden.

Programs in NextGen also evolve in name, reach, or technology. For instance, Data Communications (or "Data Comm," as it is often called) has been in development for many years under many different program names, including "Aeronautical Data Link" and "Controller-Pilot Data Link Communications" (CPDLC). Finally, the budget available to the FAA for NextGen is subject to change. When the budget for NextGen is cut, some programs have to be delayed. If budget cuts are sustained and deep enough, programs may be canceled. This budget variability affects NextGen plans and programs each year.[17]

As shown in the simplified schematic of Figure 2-1, NextGen represents a complex set of interrelated programs that require community-wide collaboration to achieve success. The next section summarizes the four groups of programs—the major players in NextGen implementation—that affect airports most directly.

[16]Advanced controller monitoring displays are high-resolution color monitors with alert algorithms (e.g., a Final Monitor Aid or FMA).

[17]Top Management Challenges for Fiscal Year 2015, Office of the Inspector General, Department of Transportation, Nov. 2014.

Copyright National Academy of Sciences. All rights reserved.

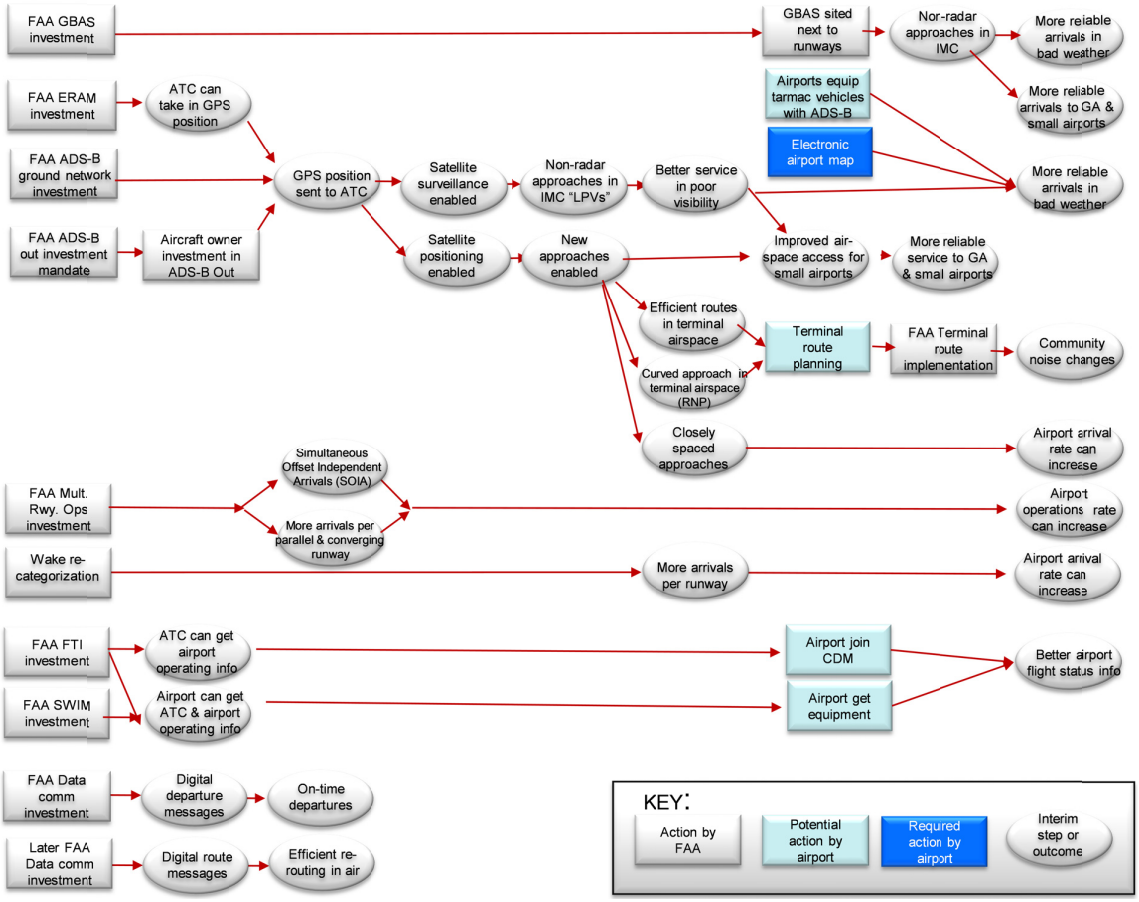


Figure 2-1. *NextGen is a set of complex, interrelated programs that require pilot, aircraft owner, airport, community, and FAA collaboration.*

Major NextGen Programs

NextGen consists of nearly 100 different modernization programs.[18] To identify the NextGen capabilities that render the highest value to pilots, aircraft owners, and travelers, the FAA established the NAC—a federal advisory committee consisting of aviation industry and FAA experts, overseen by standards-making body RTCA, Inc., to review FAA modernization plans and recommend priorities for development and implementation. The committee recommended the following four major NextGen priority focus areas.

PBN,

Surface Operations and Data Sharing (SWIM),

MRO, and

Data Comm.

[18]For current NextGen programs, brief descriptions, and the expected time frame for implementation, see FAA, NextGen, www.faa.gov/nextgen/.

Copyright National Academy of Sciences. All rights reserved.

The RTCA Committee's recommendations are documented in the *RTCA NextGen Integration Working Group Final Report, October 2014*. The FAA is currently implementing the RTCA recommendations. The FAA *NextGen Priorities Joint Implementation Plan, Executive Report to Congress, 2014*, describes the technologies being pursued, and the FAA *NextGen Priorities October 2015 Joint Implementation Plan*, Revision 1 contains status information. The FAA *NextGen Implementation Plan 2015* addresses the priority areas plus other key NextGen programs. All these reports are available online for review and download.

The following websites provide useful information on the priority programs.

NAC_Description (https://www.faa.gov/nextgen/update/collaboration/advisory_committee/) contains description of the NAC, descriptions of the priority groups and links to relevant documents

Priority_Snapshots (https://www.faa.gov/nextgen/snapshots/priorities/) contains current updates on the four focus areas.

*Priority_Completio*n (V) contains information on program completion history.

NextGen_Airports (https://www.faa.gov/nextgen/update/operator_investments_and_airports/airport_enhancements/) includes information specifically focused on airports.

This chapter describes the four priority focus areas and their potential impacts on airports. FAA provides extensive online information describing NextGen programs and their progress, including programs not in the four priority focus areas, but the information is often distributed among several websites. For example, the website NextGen_Progress_&_Plans (https://www.faa.gov/nextgen/update/progress_and_plans/) includes links for the priority programs PBN and SWIM plus enabling technologies such as ADS-B and NAS Voice System, but does not include MRO. A web search for MRO uncovers additional sites, such as MRO_Portfolio (https://www.faa.gov/nextgen/snapshots/portfolios/?portfolioId=9) and MRO_Snapshot (https://www.faa.gov/nextgen/snapshots/priorities/?area=mro), containing different types of information on technical content, schedules, and progress. An interested researcher needs to examine several sites and associated links to gain a full picture of a particular program or technology. This guide provides basic information about each program and suggests links.

PBN

A key enabler for NextGen is the shift from solely ground-based radar and navigation aids to include the use of navigation data based on the Global Navigation Satellite System (GNSS). GNSS includes position and timing signals from U.S. GPS satellites, error correction/accuracy improvement to GPS signal provided by WAAS and local area ground-based augmentation systems (GBAS), and aircraft GNSS receivers including signal integrity monitoring. The use of GNSS increases both navigation accuracy and flexibility. PBN refers to en route and terminal operations that take advantage of the improved navigation performance available from automation using either or both space-based or ground-based sources. As described by the FAA,

> Performance-Based Navigation (PBN) is comprised of Area Navigation (RNAV) and Required Navigation Performance (RNP) and describes an aircraft's capability to navigate using performance standards.
>
> RNAV enables aircraft to fly on any desired flight path within the coverage of ground or space-based navigation aids, within the limits of the capability of the self-contained systems, or a combination of both capabilities.
>
> RNP is RNAV with the addition of onboard performance monitoring and alerting capability. A defining characteristic of RNP operations is the ability of the aircraft navigation system to monitor the navigation performance

Copyright National Academy of Sciences. All rights reserved.

it achieves and inform the crew if the requirement is not met during an operation. The performance requirements of PBN are conveyed to the operators through navigation specifications. PBN navigation specifications include Advanced RNP (A-RNP), RNP 0.3, RNP 1, RNP 2, RNP 4, RNAV 1, RNAV 2, RNP 10 (RNAV 10), as well as RNAV (GPS) and RNAV (RNP) approaches.[19]

In the above description, RNAV 1 and RNP 1 refer to navigation performance criteria where the aircraft will be within 1 nautical mile of the assigned track with 95% confidence. The other RNAV and RNP values represent the 95% confidence for their respective nautical mile performance criteria. RNAV/RNP values of 4, 5, and 10 apply to oceanic en route operations; values of 1 and 2 refer to domestic en route and terminal area operations, and values of 0.3 and 0.1 apply to terminal instrument approach procedures. To understand the benefits of PBN, it helps to understand air traffic management history.

Navigation History

En Route Phase

Aircraft navigation across the United States has evolved from dead reckoning with compass, map, stars, and visual landmarks, first based on revolutions in radio, and more recently, on advances in aircraft electronics (avionics), digital computers, and space-based navigation systems. Bonfires and rooftop arrow markers were early ground references, replaced in the early-20th century first with radio direction finders that homed in on radio stations, and then with dedicated aviation, non-directional, radio beacons and very high frequency (VHF), directional, radio beacons. The directional VHF omnidirectional range (VOR) radio beacons emit signals that identify the beacon, and generate radio signals that transmit radial rays around the beacon. The aircraft VOR radio receiver decodes the signal and displays the directional radial from an emitter. Crossing two radials from two different VORs (with two radios) allows precise lateral position determination by triangulation. The other current primary radio navigation element is the VHF distance measuring equipment (DME), which is often co-located with a VOR. The DME provides precisely timed responses to aircraft interrogations enabling determination of slant range between the aircraft and the DME location. With range and bearing data, aircraft position can be determined with single VOR/DME radio and site. The non-directional and directional radio beacon systems are collectively referred to as navigational aids (NAVAIDs). Use of these NAVAIDs by the aircraft required only relatively simple aircraft radio equipment. Navigation "fixes" could be defined by either intersections of VOR radials or the physical location of individual NAVAIDs. Published flight routes based on these fixes, often called "highways in the sky," are used to manage air traffic.

Flight plans listing the names of NAVAIDs as waypoints along the flight plan are loaded in aircraft flight management computers and filed with ATC. Flight plans can be altered during flight, particularly to avoid weather, turbulence, and adverse winds. NAVAIDs are often about 50 miles apart to maintain good frequency reception.

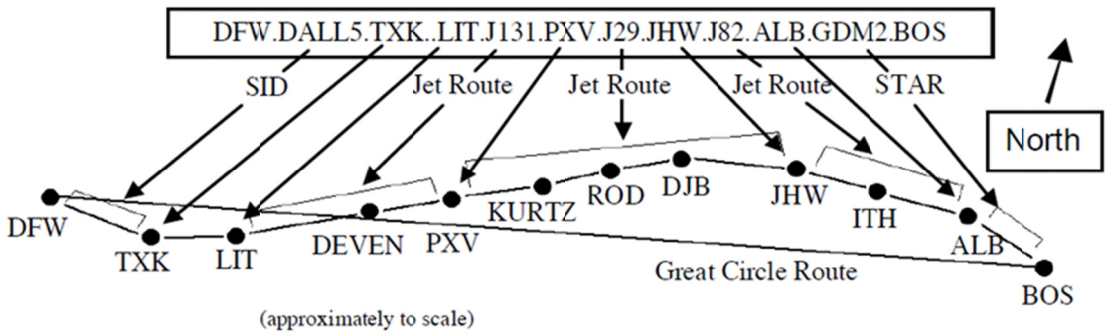
Navigation based on flying from one geographically fixed NAVAID to another provided a major safety improvement but has disadvantages. First, following the zigzags from NAVAID to NAVAID adds time and distance to a route as shown in Figure 2-2, increasing fuel costs. Second, the limited number of routes can become congested due to traffic or weather. On the other hand, a limited list of fixed NAVAID waypoints can be more easily memorized by pilots and controllers, enabling them to maintain a mental picture of the traffic in their airspace.

Modern aircraft navigation systems with RNAV avionics are capable of navigating with virtual waypoints defined by relative distances from ground NAVAIDs or on GPS-based latitude and longitude. These RNAV routes can be constructed to reduce distances, spread traffic demand, or follow more efficient routes. The FAA has established RNAV routes defined by NAVAID intersections and distances.

[19]FAA Performance Based Flight Systems Branch PBN Information sheet, https://www.faa.gov/about/office_org/headquarters/avs/offices/afs/afs400/afs470/pbn/.

Copyright National Academy of Sciences. All rights reserved.

RNAV avionics also allows use of "direct" routes where the aircraft can fly from an arbitrary staring point to a real or virtual waypoint.



Source: Erzberger, Heinz, and McNully, David. 2001. Method and System for an Automated Tool for En Route Traffic Controllers. U.S. Patent 6,314,362 B1, filed February 2, 2000, and issued November 6, 2001.

Figure 2-2. *Typical NAVAIDs-based routes are circuitous.*

The use of RNAV operations enabled by modern avionics has been supported by upgrades to the air traffic management computers and displays. ATC computers and displays could not accept GPS-based latitude and longitude position reports until the en route automation modernization (ERAM) computer hardware and software upgrades were accomplished nationwide. On the human side, NextGen ATC decision support tools are intended to aid the human controller in maintaining situational awareness with the more complex en route airspace.

Transition to Terminal Airspace and Airport

Leaving cruise, a pilot executes a planned descent to the airport airspace. The descent from cruise—a controlled and planned change in altitude—dissipates both speed and altitude; if done efficiently, it is smooth and burns very little fuel. Due to traffic and spacing requirements, current approaches are often not efficient and involve radar vectoring instructions for both step downs with level flight segments and lateral excursions, both of which add time and burn extra fuel. A major goal of the FAA is to use NextGen PBN technologies to support more of the efficient optimized profile descents (OPDs).

In order to manage arrival traffic in high traffic airspace, to minimize communication errors, and to reduce communication workload, the FAA has produced for major airports standard terminal arrival routes (STARs) which prescribe a flight route defined by a series of NAVAIDs extending from the last en route fix, approximately 100 miles from an airport, to a terminal airspace fix in the vicinity of the airport. Because aircraft approach from many directions, each airport has many STARs. The basic Standard Terminal Automation Replacement System (STARS) is based on aircraft navigation using standard NAVAIDS, and the resulting aircraft position accuracy limits the number of STARs and waypoints one can have in a given section of airspace. Sometimes, several airports in the same vicinity share a STAR; aircraft follow the same arrival path up until the final waypoint when they join the airport approach. This can result in congestion, with delays propagated across nearby airports. Figure 2-3 is an example STAR for the Dallas-Ft. Worth Metroplex that terminates at the FINGR fix and serves several airports.

Along with increased flexibility in establishing waypoint location, RNAV/RNP flight precision enables establishing multiple closely spaced flight paths, allowing individual and non-overlapping STARs for airports in the same vicinity. Figure 2-4 shows an RNAV STAR in the Dallas-Ft. Worth Metroplex. Using RNAV/RNP technology with more flexible waypoints along an approach enables shortening flight

Copyright National Academy of Sciences. All rights reserved.

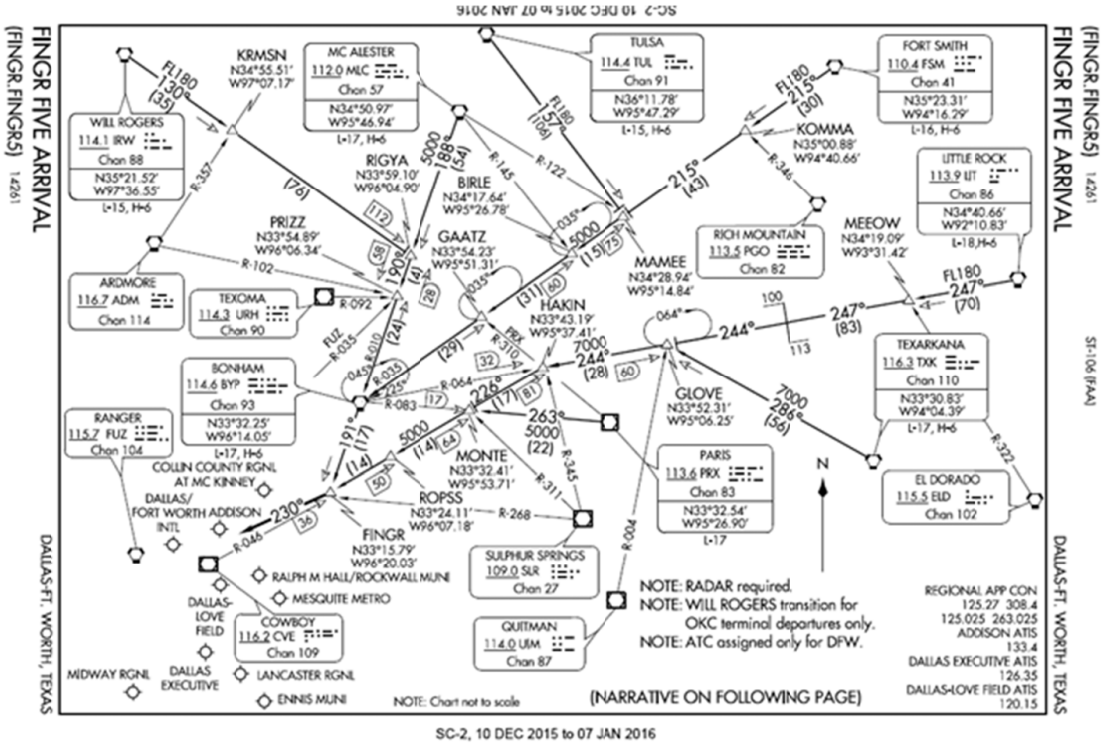


Figure 2-3. *Conventional STAR for Dallas-Ft. Worth.*

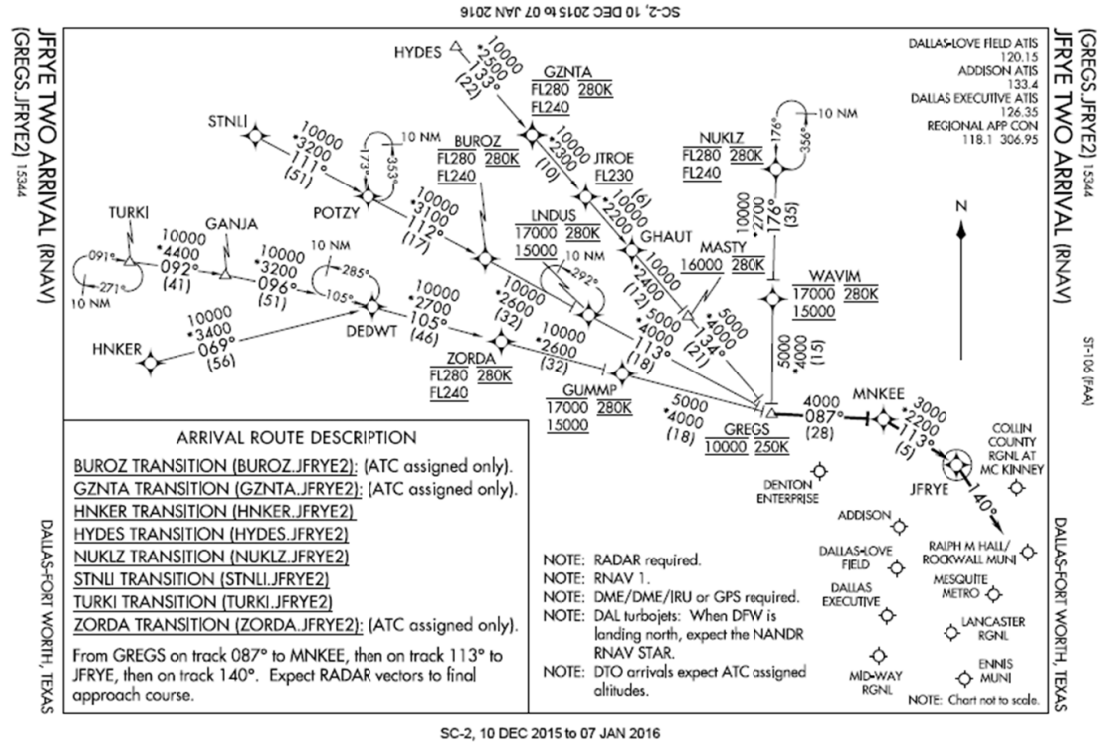


Figure 2-4. *RNAV STAR for Dallas-Ft. Worth.*

Copyright National Academy of Sciences. All rights reserved.

routes to reduce time and fuel burn and also enables an increase in the potential number of flight routes.

Approach and Landing Phase

In the approach airspace, controllers vector the arrivals to intercept the final approach path or to a charted arrival procedure. A charted procedure consists of another set of NAVAID-based fixes with specified lateral and altitude requirements leading to interception of the final approach path. The ILS final approach consists of the localizer (lateral) and glide slope (elevation) radio beacons, which provide precision azimuth and elevation guidance to the runway. Because ILS approaches are based on radio beams, they require relatively long obstacle-free, straight arrival paths for aircraft to get established on the localizer course, and cannot be used when terrain or other obstructions do not permit such long paths. The ILS produced a revolutionary improvement in low-visibility operations; however, it only supports straight line approaches and requires several miles of clear distance leading to the runway, and the cost precludes implementing ILSs at smaller airports, and the required straight path precludes use where terrain does not allow a clear straight approach.

NextGen RNAV/RNP technology supports efficient, continuous routes through the terminal airspace to the final approach, and can include curved flight routes to avoid terrain obstacles Figures 2-5a and 2-5b compare the basic ILS and the RNAV/RNP approaches to Runway 31L at Dallas Love Field.

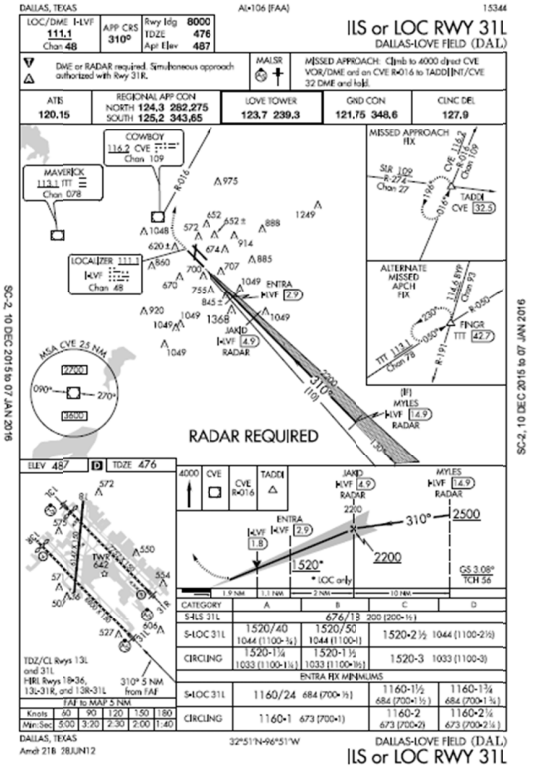


Figure 2-5a. *DAL, Rwy 31L ILS approach.*

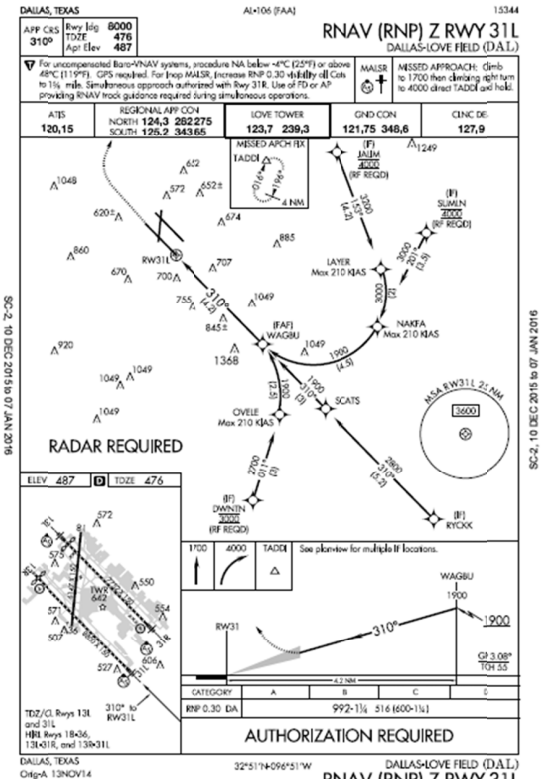


Figure 2-5b. *DAL, Rwy 31L curving RNAV/RNP approach.*

Copyright National Academy of Sciences. All rights reserved.

Raw GPS data are not sufficiently accurate to support low-visibility RNAV/RNP instrument approach procedures. The WAAS and the GBAS both provide corrections to the GPS position data. The WAAS corrections cover the entire continental U.S. and are adequate for near-ILS Category I approaches. The GBAS corrections are local and are expected to potentially support ILS Category III level approaches. The FAA work program is now focused on validating standards for a GBAS Approach Service Type–D (GAST-D) (Category-III minima) service. The program projects that a GAST-D can be available by 2018. Both straight and curved PBN approaches can be supported by WAAS and GBAS, allowing their use with terrain-challenged runways.

GBAS has several advantages over traditional ILS. One GBAS station can support multiple runway ends and reduce the total number of systems at an airport. This reduces the radio frequency requirements and simplifies airport infrastructure. Unlike ILS, which requires one frequency per runway, a GBAS requires only one frequency assignment for up to 48 individual approach procedures. GBAS has more flexible siting criteria, enabling it to serve runways that cannot support an ILS because of insufficient space for conflict-free localizer and glide slope critical areas. A GBAS is sited to minimize NAVAID critical areas, placing fewer restrictions on aircraft movement during ground taxi and air operations. The GBAS approach guidance is steadier than ILS approach guidance. Also, GBAS requires less frequent flight inspections than those required of ILSs. Despite these advantages, GBAS, so far, has only been applied to a "niche market," because the majority of airports requiring Category II and III landing minimums already have adequate ILSs installed.

Departures and Climb Out

Standard instrument departures (SIDs) are charted departure paths analogous to STARs. Like STARs, their use reduces workload and communication requirements, and, as with STARs, the use of an SID can create a delay as aircraft wait for the pathway out of an airport to be clear before taking off. NextGen RNAV/RNP flight precision supports procedures to alleviate this delay by allowing more closely spaced and spatially separated SIDs. Spatially separated SIDs can allow non-airline operators to depart an airport near a major airline hub airport when previously the non-airline operators (such as business aircraft operators and GA) would have faced a delay.

New SIDs can introduce noise to new areas, or re-distribute aircraft noise among the paths. Aircraft using RNAV/RNP SIDs have much more accurate navigation performance, and follow the defined SID much more closely than with conventional SIDs, which can concentrate flight paths and increase aircraft noise level under an existing departure route. Without sufficient advanced planning and community outreach, these changes can cause significant public controversy and increased noise complaints even with the implementation of single new RNAV/RNP SID.

PBN Terminal Area Summary

The FAA has bundled PBN implementation together for sets of airports. The FAA program name is Optimization of Airspace and Procedures in the Metroplex, or OAPM, but it is also commonly called Metroplex. The FAA PBN website, PBN_Progress_&_Plans (https://www.faa.gov/nextgen/update/progress_and_plans/pbn/), contains a description of PBN, and links to information on Metroplex and single airport RNAV/RNP programs.

Adjusting arrival and departure routes by even a few tenths of a mile can mean aircraft traffic, and noise, is introduced over a new area near an airport. This is particularly noticeable when the new pattern affects airport departures. Taking off and climbing uses most of an aircraft's thrust, so the aircraft's engine or engines are operating at a high pitch and generating the loudest noise of the entire flight.

Copyright National Academy of Sciences. All rights reserved.

Airports for which noise is an issue should know that they may be able to influence the placement of new RNP routes and NextGen Metroplex routes if they engage in the planning process before a new route is finalized. Both RNAV SIDs and RNAV STARs can be designed to require an aircraft to operate over non-noise-sensitive areas and avoid noise-sensitive areas. Identifying the noise-sensitive areas and ensuring that RNAV/RNP flight routes are not placed over them would reduce subsequent aircraft noise complaints.

More extensive and detailed discussion of PBN can be found in *ACRP Report 150: NextGen for Airports, Volume 1: Understanding the Airport's Role in Performance-Based Navigation: Resource Guide*, ACRP Report 03-34.

Surface Operations and Data Sharing

The goal of the improved surface operations and data sharing is to increase efficiency and reduce aircraft delay through improved situational awareness and collaborative decision making (CDM). Airport operators could play a significant role in the implementation of both situational awareness and CDM.

The FAA has published a concept of operations document (ConOps) for surface CDM which states:

> Surface CDM is the sharing of flight movement and related operational information among Airport Operators, Flight Operators, Flight Service Providers, and FAA Stakeholders to improve demand and capacity predictions, thereby enabling those who practice the Surface CDM concept to maximize the use of available airport and airspace capacity, while minimizing adverse effects on Stakeholders, passengers, and the environment.[20]

The ConOps includes situational awareness-related data requirements for airport operators including runway status, ramp operations, deicing operations, and security.

Under the NextGen program the FAA is developing the Terminal Flight Data Manager (TFDM) system, which will enable greater sharing of data among controllers, aircraft operators, and airports so they can better stage arrivals and departures to improve efficiency in the terminal. TFDM operates through four functions:[21]

- Electronic flight data distribution,
- Traffic flow management,
- CDM on the airport surface, and
- Systems consolidation.

The near-term priority efforts are directed to converting to electronic flight strips under the Advanced Electronic Flight Strips (AEFS) program and sharing of surface traffic data among the towers and terminal radar approach control (TRACONs). The SWIM system is the backbone of the data distribution system. SWIM is designed to provide the single point of access supported by standardized data formats for near real-time access to NAS data. SWIM will include a broad array of aeronautical, flight, and weather information. In accordance with the NAC surface operations priority, early SWIM implementations have included surface visualization tool (SVT) and the SWIM Terminal Distribution System (STDDS) that supports airport surface data distribution to both the airport towers and TRACON, and the SWIM SVT that displays the terminal surface data in the Tower and TRACON.

[20]U.S. Airport Surface Collaborative Decision Making (CDM) Concept of Operations in the Near-Term, FAA Surface Operations Office, June 15, 2012, p1-2.

[21]NextGen Decision Support Systems at NextGen_Decision_Support_Systems (https://www.faa.gov/nextgen/update/progress_and_plans/decision_support_systems/#tfdm).

Copyright National Academy of Sciences. All rights reserved.

General SWIM information and status can be seen at the FAA website "System Wide Information Management" SWIM_Progress_&_Plans (http://www.faa.gov/nextgen/update/progress_and_plans/swim/). The websites Surface_Ops_Progess_&_Plans (https://www.faa.gov/nextgen/snapshots/portfolios/?portfolioId=7) and Surface_Ops_Priorities (https://www.faa.gov/nextgen/snapshots/priorities/?area=sops) contain general and status information on the surface operations. The website https://www.faa.gov/about/office_org/headquarters_offices/ato/service_units/pmo/proj/tfdm/ provides information on the goals and status of TFDM.

The decisions on surface data access are made by a CDM stakeholders group (CSG). As noted in the 2014 NextGen Priorities Joint Implementation Plan:

> The [Collaborative Decision Making Stakeholders Group] CSG provides oversight and governance of joint FAA/industry CDM initiatives and provides the FAA with input on prioritization and tasking for possible technologies, tools, and/or procedures that will increase the efficiency of the NAS. The CSG includes representatives from the Airlines for America, National Business Aviation Association (NBAA), Regional Airline Association (RAA) and the current industry lead of the CDM program, Delta Air Lines.[22]

The October 2015 revision to the plan notes that:

> Several milestones were completed relative to the two-way data sharing agreement negotiated by the Collaborative Decision Making Stakeholders Group (CSG). First, the CSG reached agreement to allow airport operators to participate in data exchange collaborative decision making related processes and procedures. Like flight operators, airports require real-time air traffic control and flight movement information to manage airside operations more effectively. This is especially true for airports that provide ramp control in the non-movement areas. More uses of real-time information include better gate management, forecasting of airport resource demands and preparing for irregular operations such as severe weather operations.[23]

Airport operators can get access to SWIM data through the NAS Enterprise Messaging System (NEMS). The data products available on SWIM and links to instructions on getting access to SWIM data are available at the FAA website "Data Products Available via SWIM," https://www.faa.gov/nextgen/programs/swim/products/.

Surface traffic surveillance data are available at the 35 airports that are equipped with airport surface detection equipment—Model X (ASDE-X) and at the nine airports that are equipped with airport surface surveillance capability (ASSC). Data are provided by the ASDE-X and, for ASSC, the ASDE-3 radar supplemented by multilateration location information based on aircraft transponder responses. In the future ADS-B data may be used to provide more accurate surveillance data at additional airports.

MRO

As mentioned at the beginning of the chapter, MRO are included in NextGen, but are not based primarily on the use of improved navigation, surveillance, and communications. Rather they are based primarily on analysis aircraft performance data and analysis of aircraft wake generation and wake upset susceptibility data. Most of the MRO technologies apply to parallel runway operations; however, NextGen Wake RECAT, while not restricted to MRO, is included programmatically under MRO. The applicability of MRO technologies to an airport is a function of the runway configuration for parallel operations and the traffic weight class mix for Wake RECAT.

Wake RECAT redefines the FAA aircraft wake vortex class structure. Current classes, including Small, Large, Boeing 757 (B757), Heavy and Super [Heavy] are based on weight only. Wake RECAT defines six classes, A–F, based on weight and wingspan to better reflect leader wake hazard and follower wake upset susceptibility. Wake RECAT classes and separation requirements are defined in FAA Order JO

[22]NextGen Priorities Joint Implementation Plan, 2014.
[23]NextGen Priorities Joint Implementation Plan, Revision 1, October 2015.

Copyright National Academy of Sciences. All rights reserved.

7110.659C, *Wake Turbulence Recategorization*. Practically, RECAT removes the additional wake vortex separation requirement that existed for Large class aircraft following Boeing 757 aircraft, and reduces the separation requirements for certain Heavy class aircraft following smaller Heavy class aircraft. It has the biggest impact at international and cargo airports that have significant heavy and B757 traffic. The removal of the additional separation requirement behind the B757 has been included in FAA Order JO 7110.65W as a general change independent of Wake RECAT. The use of full Wake RECAT with the new classes has been implemented at specific airports, including Memphis (MEM), Louisville (SDF), Atlanta (ATL), Houston (IAD), Cincinnati (CVG), Chicago (ORD), Charlotte (CLT), Newark (EWR), and New York (JFK and LGA).

NextGen includes four significant impacts on parallel runway operations in instrument conditions. The first is the reduction of the minimum runway centerline separation for dual simultaneous independent approaches from 4,300 feet to 3,600 feet (3,000 feet with a 2.5–3.0 degree offset approach) without high update radar but with advanced controller monitoring displays. The second is the reduction of the minimum runway centerline separation for triple simultaneous independent approaches from 5,000 feet (4,300 feet with advanced controller monitoring displays) to 3,900 feet (3,000 feet with 2.5-3.0 degree offset approaches to both outside runways) without high update radar but with advanced controller monitoring displays. The third is the reduction of minimum diagonal separation for arrivals on parallel runways separated by 2,500 to 3,600 feet from 1.5 nautical miles to 1.0 nautical mile. These changes, based on analysis of aircraft performance, are generally applicable and are already included in FAA 7110.65W. The fourth is the ability to use dependent (staggered) approaches with runway centerline separations from 2,500 feet to as low as 750 feet for specific airports and specific classes of aircraft. This procedure is airport specific, based on wind and wake data, and is governed by FAA JO 7110.308A. These improvements can have impact on airport runway construction options.

NextGen wake turbulence mitigation refers to the use of wind data and sensing to determine if operations on an upwind runway of a closely spaced pair can be operated independently from the downwind runway. This is especially important for departures where extended wake separation requirements on either runway affect operations on both. This wake turbulence mitigation for departures (WTMD) technology is most likely to provide benefits for airport such as SFO where there are persistent crosswinds on the departure runways. The most recent FAA position is that WTMD will remain at SFO for periodic testing, but will not be deployed elsewhere.

In addition to the above, the need to increase capacity on very closely spaced parallel runways has resulted in the development of offset approaches. Under these procedures, aircraft conduct independent approaches as if they will land on widely spaced parallel runways, but one aircraft must be lined up on an offset angle to its actual runway centerline. During the final part of approach, the offset aircraft establishes visual contact with the runway and parallel traffic and moves inward to the real runway extended centerline.[24]

Data Comm

Data Comm services will provide digital communications services between pilots and air traffic controllers (Figure 2-6), as well as enhanced ATC information to airline operations centers. The capabilities will enhance safety by reducing communication errors, increase controller productivity, and increase airspace capacity and efficiency while reducing delays, fuel burn, and carbon emissions at 56 towers nationwide.

[24]San Francisco approach RNAV(PS) PRM X RWY 28R (Simultaneous Close Parallel) Terminal Approach Plate.

Copyright National Academy of Sciences. All rights reserved.



Figure 2-6. *Typical Data Comm display shows pre-departure clearance from tower.*

Pre-departure clearance allows tower controllers to give pilots complex taxi instructions in a text format. At large airports with complex taxiway layouts, having text displays of the routes by taxiway name decreases the possibility of error in taxiing and increases safety. Previously, pilots often wrote down taxi instructions as they were given and followed their handwriting.

Pre-departure clearances can be granted using the commercial Aircraft Communication Addressing and Reporting System (ACARS). A different medium will be used for en route Data Comm. Air traffic controllers do not have ACARS terminals for transmitting ACARS-based messages. For this application, the FAA will use Future Air Navigation Systems (FANS) 1/A+ avionics and VHF Data Link Mode 2 (VDL-2) radios for en route Data Comm. FANS 1/A has been in use over the ocean for many years, so many large aircraft already have this equipment.

The FAA has established NextGen equipage funds to provide interest-free financing for avionics equipage, prior to ground station implementation of the technology. Both the controller and the pilot need to have Data Comm equipment to realize the Data Comm benefits, so the FAA set up an incentive fund to encourage the aircraft side of the Data Comm investment. Data Comm will provide a means of two-way communication for ATC clearances, instructions, advisories, flight crew requests, and reports.

Data Comm is critical to the success of NextGen, enabling efficiencies not possible with the current voice system. En route services will include airborne reroutes, controller- and pilot-initiated downlinks, altitude and altimeter settings, tailored arrivals, and issuing of crossing and holding restrictions, advisory messages, beacon codes, transfer of communications, and initial check-in.

Data Comm implementation is scheduled in two stages: initial services (such as pre-departure clearance) and a much fuller implementation called full services. The Data Comm en route services will contribute to enhanced safety, more efficient routes, and a reduction in flight delays, resulting in lower fuel burn for operators.

In accordance with NAC priorities, the FAA is planning to install Data Comm at 56 Airport Traffic Control Towers by the end of 2016. Data on the installation status can be found at the Data_Comm_Progress_&_Plans (https://www.faa.gov/nextgen/update/progress_and_plans/data_comm/).

Aircraft Equipment Mandate

The FAA has mandated that every aircraft seeking access to Class A, B, and C airspace, and Class E airspace above 10,000 feet, must be equipped with GPS-capable ADS-B equipment that transmits a

Copyright National Academy of Sciences. All rights reserved.

GPS-based position of the aircraft, by January 1, 2020. Broadcasting the aircraft's position is commonly referred to as "ADS-B Out," for outbound communication. For an aircraft equipped with the ability to receive the signals, receiving the broadcast is called "ADS-B In." With both ADS-B Out and ADS-B In, suitably equipped aircraft can view all transmitting aircraft relative to their position on a cockpit display. In is not mandated; only Out is mandated. Aircraft (usually GA) equipped with traffic information system–broadcast (TIS-B) will also be able to see the position of nearby aircraft.

ADS-B Out is mandated because universal equipage provides the ability to track the position of all aircraft more accurately and independently of radar. The FAA explained that aircraft owners who saw the benefit of ADS-B In could choose to acquire it.

The ADS-B Out mandate is triggering interest from GA aircraft owners in a broader range of avionics upgrades. The NextGen GA Fund was established to finance these installations, including WAAS–capable GPS, ADS-B (In and Out), RNAV/RNP avionics, transponders, Data Comm, SWIM displays, instrument panel modifications, antennas, other enabling electronic components, installation, and certification costs. Pilots operating under *Federal Aviation Regulation* Parts 23, 91, and 135 are likely to be early adopters. Pilots and aircraft owners interested in the fund provisions should contact NEXA Capital Partners, LLC. Up-to-date information, including information on a $500 ADS-B rebate for single-engine aircraft, can be found at the FAA NextGen Equip ADS-B website, Equip_ADS-B (http://www.faa.gov/nextgen/equipadsb/). Aircraft will lose access to certain U.S. airspace without timely equipage.

In 2015, there were over 200,000 GA aircraft registered in the United States. Over 120,000 were single-engine piston, fixed-wing aircraft.[25] Many are more than 40 years old. About 6,000 of these aircraft had been equipped with ADS-B Out as of July 2014.[26] Equipping the entire fleet would require nearly 300 new installs per week until the deadline. Equipping to meet the mandate costs from $1,000 to $7,000 per aircraft, depending on how high the aircraft flies and how many functions the owner chooses. Not all GA aircraft will equip; some will choose to remain clear of the airspace requiring ADS-B instead.

The cost and trouble of equipping aircraft to enable new services is one of the costs of NextGen, and some aircraft owners are not happy about the mandates. Also, the choice of equipment depends on what the aircraft owner wants to do with the new capabilities, and sometimes that is not straightforward. Changing deadlines and changing program requirements at FAA in the past have made some users wary of NextGen promises and schedules,[27] even though NextGen offers the benefits of fuel efficiency and reliable flight service.

Geographic Information System (GIS) Mandate and Airport Improvement Program (AIP)

The FAA Memorandum, "Airports Geographic Information System (GIS) Transition Policy for Non-Safety Critical Projects," dated August 23, 2012, mandated immediate incorporation of safety-critical data into the GIS for all airports in the National Plan of Integrated Airport Systems (NPIAS) and phased incorporation of non-safety-critical data for specific airport classes in the NPIAS.[28] Cost is recognized as a major issue; use of AIP funding is permitted for above-ground surveys. Several NextGen technologies specifically depend on airport geospatial data.

Federal funding for specific airport improvements is available under the FAA AIP. Extensive information on the AIP, including the 3,345 eligible airports, types of projects, contacts, and much more, is

[25]FAA General Aviation and Part 135 Activity Surveys – CY 2014, Table 1.1.
[26]John Croft, "ADS-B Options Increase As Operators Resolve To Equip," *Aviation Week*, Jul 30, 2014.
[27]Bill Carey, "American Airlines Awaits Fruit of NextGen Investment," *Aviation International News*, September 16, 2013.
[28]FAA_GIS_policy_memo (http://www.faa.gov/airports/planning_capacity/airports_gis_electronic_alp/media/airportsGISTransitionPolicy.pdf).

Copyright National Academy of Sciences. All rights reserved.

available at the FAA website Airports_AIP (http://www.faa.gov/airports/aip/). AIP funding for GIS is addressed in the AIP handbook, FAA JO 5100.38D paragraph 3-82,[29] and GIS-related information is on the following FAA websites.

- FAA_Implementation_Guidance_for_Airports_GIS_and_eALP (http://www.faa.gov/airports/planning_capacity/airports_gis_electronic_alp/).
- FAA_GIS_Advisory_Circulars (http://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.list/parentTopicID/187).
- FAA_Airports_GIS_Survey (https://airports-gis.faa.gov/public/airportsSteps.html).
- FAA_Airports_GIS_Public (https://airports-gis.faa.gov/public/).
- FAA_GIS_Info (http://airports-gis.info/).
- NPIAS_airports (http://www.faa.gov/airports/planning_capacity/npias/).
- Part_139_airports (http://www.faa.gov/airports/airport_safety/part139_cert/).

Much more information on GIS is included in *ACRP Report 150: NextGen for Airports, Volume 4: Leveraging NextGen Spatial Data for Airports: Guidebook*.

[29]FAA JO 5100.38D, September, 30, 2014, http://www.faa.gov/airports/aip/aip_handbook/.

Copyright National Academy of Sciences. All rights reserved.

3 NextGen Programs and Portfolios

This chapter presents tabular fact sheets about major NextGen programs and program portfolios, with an analysis of each program's impacts on airports, from the present to 5 to 10 years out. The FAA distinguishes between programs and program portfolios: a NextGen program is a distinct project that either provides critical NextGen capabilities or the infrastructure on which critical NextGen capabilities will be built. A NextGen portfolio is a set of related programs, including procedures, that incrementally transform operation of the national airspace and airport system. In addition, this chapter describes six near-term NextGen technologies that are part of the programs and portfolios. Each of these technologies is an important part of NextGen capabilities infrastructure, and is often discussed in isolation.

This presentation identifies key aspects of each major NextGen program, portfolio, or technology:

What the names of the NextGen programs are.

Why a particular program is being undertaken, such as what it fixes or improves.

How the program is being implemented, in terms of procedures and equipment.

Who is implementing it, and whom it will affect?

Where any physical equipment will be installed, where procedures will be changed, and which airports will be affected.

When the program will be implemented, and separately, when airports will notice an impact or when action will be required. The time frames are near term (2015–2018), midterm (2019–2022), and far term (2022 onward). When available, the exact schedule is reproduced.

The tables also identify the effect on airports, actions that airports can take, whether it belongs to a NextGen prioritized major program, and where to find more information.

Copyright National Academy of Sciences. All rights reserved.

The chapter includes fact sheets on the following programs, portfolios, and technologies:

NextGen programs:

- ADS-B Out.
- DataComm.
- ERAM.
- Terminal Automation Modernization and Replacement (TAMR).
- SWIM.
- NAS Voice System (NVS).

NextGen portfolios:

- Improved Surface Operations.
- Improved Approaches and Low-Visibility Operations.
- Improved MRO.
  - 1: Closely Spaced Parallel Runways Procedures.
  - 2: Wake Turbulence Mitigations.
- PBN.
- Time-Based Flow management (TBFM).
- Collaborative Air Traffic Management.
- Separation Management.
  - 1: Wake RECAT.
  - 2: Oceanic In-Trail Procedure.
- On-Demand NAS Information.
- Environment and Energy.
- System Safety Management.
- NAS Infrastructure.

Related technologies:

- Cockpit Display of Traffic Information (CDTI).
- WAAS.
- GBAS.
- WAM.
- Flight-Deck Interval Management (FIM).
- Surface Management.
- Airport Geographical Information System (GIS).

Copyright National Academy of Sciences. All rights reserved.

What	**Automatic Dependent Surveillance-Broadcast Out (ADS-B Out)**
Why	ADS-B Out is the NextGen surveillance component that requires aircraft to broadcast identity, position, and velocity, and other data that are picked up by ground stations and transmitted to controllers. ADS-B Out 1-sec update-rate and GPS-based flight data are major improvement over radar. Also, ADS-B provides essentially universal geographic coverage and provides a major surveillance improvement in areas where radar coverage is spotty or non-existent.
How	ADS-B Out consists of aircraft broadcasting their GPS-based position and speed, along with barometric altitude and aircraft ID, every 1–2 seconds. This provides higher accuracy, broader geographic coverage, and higher update rates than current radar surveillance. ADS-B Out equipment has aircraft and ground infrastructure components.
Who	Commercial jetliners must carry ADS-B Out–capable Mode S transponders to transmit their position. A lower-cost universal access transceiver (UAT) radio is available to GA aircraft.
Where	Ground-based ADS-B is in place throughout the United States and the Gulf of Mexico.
When	Ground-based ADS-B Out equipment is in place now. The FAA has mandated that all aircraft operating in controlled airspace must have ADS-B Out by January 1, 2020.
Effect on Airport	ADS-B Out can potentially provide data for ground tracking surveillance at airports not having ASDE-X, and increased instrument capacity at airports without radar coverage. ADS-B Out should be able to provide high update-rate precision surveillance to support instrument approaches to closely spaced parallel runways.
Action Needed by Airport	No action is needed by airports for installation, but equipage of ground vehicles with ADS-B Out is available for improved safety benefits. An FAA contract is in place to install nationwide transceivers and repeaters to share information across the several channels that ADS-B uses.
Priority Area	PBN
More Info	NextGen_ADS-B (http://www.faa.gov/nextgen/programs/adsb/) 14 CFR 91.225 ADS-B Out Equipment and Use and 14 CFR 91.227 ADS-B Out Performance Requirements both at Federal_Law (http://www.ecfr.gov/cgi-bin/text-idx?SID=adcd42bf1eae8417436c765b02aae2f8&mc=true&tpl=/ecfrbrowse/Title14/14tab_02.tpl)

Copyright National Academy of Sciences. All rights reserved.

What	**Data Communications (Data Comm)**
Why	Data Comm is the NextGen communications component that enables controllers and pilots to communicate with text messages, rather than rely solely on voice communications. This will result in fewer errors (read-back, for example) and greater safety, particularly for complex ground taxi instructions and flight rerouting. It also provides another way for controllers to communicate when the voice channel is crowded.
How	Also known as controller-pilot data link communications (CPDLC), the first application is airport data link delivery of surface departure clearances (DCLs), which is being tested at the Memphis (MEM) and Newark (EWR) airports. Other near-term applications are (1) flight rerouting negotiation and (2) traffic information service, in which aircraft can see other aircraft on a radar-like display in the cockpit.
Who	Data Comm and CPDLC require digital radios, associated communications control avionics and data displays that typically exist only on air carrier and high-end business aircraft.
Where	Rockwell Collins ARINC and SITA VHF ground stations cover the U.S. fairly well. FAA CPDLC equipment is installed on an airport-by-airport basis.
When	Pre-departure clearance and traffic information service for pilots are scheduled for the near term (2016 at 56 airports); the en route data link is slated for the far term.
Effect on Airport	In the near term, gate agents and ramp towers will communicate with an aircraft with a text message. (Radio frequency interference concerns are similar to the current ones, as these are not new band channels.) This improves efficiency in airport taxi by reducing controller voice workload. In the midterm, this could reduce workload for gate staffing. In the far term, Data Comm is one of several technologies that will improve on-time schedule reliability, especially during weather events.
Action Needed by Airport	Aircraft at the gate must be able to communicate with ATC; communications links that are lacking may need to be installed.
Priority Area	Data Comm
More Info	FAA_DataComm (http://www.faa.gov/about/office_org/headquarters_offices/ato/service_units/techops/atc_comms_services/datacomm/)



Copyright National Academy of Sciences. All rights reserved.

What	**En Route Automation Modernization (ERAM)**
Why	Host computers could accept only radar surveillance data; moving to satellite-based position reports requires a replacement.
How	This program is the replacement for the host computer. ERAM is in place in all continental en route traffic control centers. It is required to enable ADS-B, SWIM, and Data Comm.
Who	This is the largest project the FAA has undertaken in air traffic automation in recent decades and is the "infrastructure" of NextGen.
Where	New equipment has been installed in en route ATC facilities.
When	ERAM is near term.
Effect on Airport	Although airports are not directly affected, everything in NextGen depends on ERAM as a first step.
Action Needed by Airport	Airport managers should keep abreast of any developments because ERAM full functionality and completion precedes most other NextGen initiatives. As of May 2015, ERAM is fully functional at every en route center in the contiguous states.
Priority Area	PBN
More Info	FAA_ERAM (http://www.faa.gov/air_traffic/technology/eram/)

Copyright National Academy of Sciences. All rights reserved.

What	**Terminal Automation Modernization and Replacement (TAMR)**
Why	Use of updated common terminal automation computers and displays is required to meet core NextGen capabilities.
How	TAMR converts terminal ATC facilities to the STARS, enabling use of ADS-B, adding data recording, allowing individual controller preferences, and increasing reliability.
Who	FAA facilities are affected.
Where	STARS is currently at 54 facilities and is planned for 108 additional ones.
When	Now through 2020.
Effect on Airport	Although airports are not directly affected, much in NextGen depends on TAMR as an enabler.
Action Needed by Airport	No action is needed by airports.
Priority Area	PBN
More Info	FAA_TAMR (http://www.faa.gov\air_traffic\technology\tamr\)

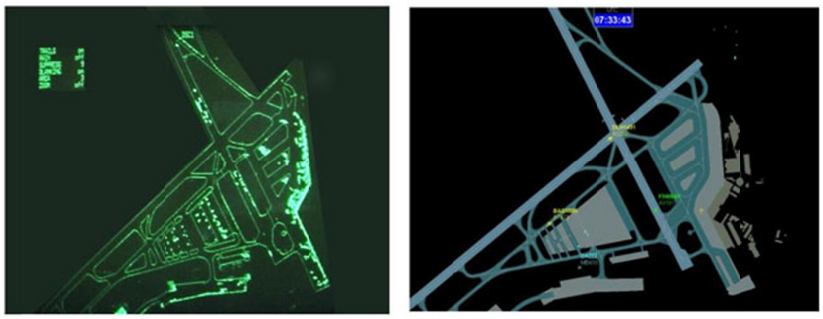


Old (Left) and new (right) STARS terminals.

(Source: www.faa.gov/air_traffic/technology/tamr/)

Copyright National Academy of Sciences. All rights reserved.

What	**System Wide Information Management (SWIM)**
Why	SWIM transforms aeronautical, airport, and weather data, previously communicated largely by voice, into a digital repository to make them available to more people in real-time and on-demand.
How	SWIM is the digital data sharing backbone of NextGen. It provides secure transmission of weather, flight, airport, and aeronautical (airspace) information among NAS operators and users. With SWIM, airport-based airport surface detection equipment, Model X (ASDE-X) surveillance data ground scheduling information, and en route flight status data are available to controllers and other airports, enabling accurate information sharing of airport impacts on related airports (such as the effect of fog or snow delays at one airport on related destination or origin airports).
Who	System wide information is available to airports, pilots, airline operational centers, air traffic personnel, military ATC, NAV CANADA, and European airspace managers for situational awareness.
Where	Portals will be established, including a web-based portal. Surveillance capability surface data are available from SFO. CLE, CVG, PIT, and MCI are planned for 2015 and 2016.
When	Sharing ASDE-X data to SWIM subscribers has begun.
Effect on Airport	Airports can use data from the system wide portal to integrate with the flight information display system and provide more reliable gate departure time information to delayed passengers. Opportunities for further development (like apps that notify passengers of wait time to departure) are likely.
Action Needed by Airport	Airports need to investigate access to SWIM data and be cognizant of airport CDM data requirements.
Priority Area	CDM
More Info	NextGen_SWIM (http://www.faa.gov/nextgen/programs/swim/) FAA_SWIM_Products (https://www.faa.gov/nextgen/programs/swim/products/)



ASDE and ASDE-X displays, which present a situational view of all aircraft on the surface and their data tags. Photo by FAA.[30]

[30]http://lessonslearned.faa.gov/ll_main.cfm?TabID=1&LLID=55&LLTypeID=10. Accessed April 15, 2015.

Copyright National Academy of Sciences. All rights reserved.

What	**NAS Voice System (NVS)**
Why	Currently, each communication to other airspaces and other controllers requires phone calls on unconnected networks, so a separate phone or network is used for each type of call. Switch replacements make it easier for controllers to talk with other users, such as airport operations centers, directly. More efficient and reliable communication supports safer operations.
How	The NVS replaces the current independent voice switches with a unified system that will enable a controller to talk with other controllers and pilots anywhere in the NAS, using the flight telecommunications infrastructure network.
Who	ATC facilities will install the NVS, which may include tower communications equipment. The maintenance responsibility is not yet specified.
Where	NVS primarily affects ATC facilities. Towers may require fiber-optic installation if they do not already have it, which is an FAA responsibility.
When	The telecommunication update is ongoing; the airport impact is in the far term.
Effect on Airport	This change enables ATC personnel to talk to more people and may someday include airport operations centers.
Action Needed by Airport	No near-term action is needed. At some point, as the SWIM is implemented, airports may be responsible for providing collaborative personnel to provide ground awareness, such as better weather information, to ease national traffic planning.
Priority Area	CDM
More Info	FAA_NVS (http://www.faa.gov/nextgen/programs/nvs/)

Copyright National Academy of Sciences. All rights reserved.

What	**Improved Surface Operations (Portfolio)**
Why	Improved Surface Operations will improve safety, efficiency, and flexibility via data exchange between controllers, pilots, and air traffic managers, to support better predictions of and adjustments to departure and arrival times. There are some places on an airport surface where tower controllers do not have good visibility, and these electronic tools provide an enhancement.
How	ADS-B position reporting and airport surface detection equipment and radar are used to better track the position and movement of aircraft on the airport surface. This program includes an allowance for airports to equip on-surface vehicles with ADS-B Out and allows for advances from enhanced flight vision systems (EFVSs).
Who	ADS-B, multilateration and transponder tracking, and primary surface radar, singly or in combination, generate information that can be used to improve surface operations and reduce potential collisions. Data feeds to airports can enhance their situational awareness.
Where	Improved Surface Operations are recommended where surface operation conflicts are a problem, where significant areas are blocked from tower or ramp control view, and where operating in low visibility is a priority.
When	In the near term to midterm, airports may acquire commercial surface tracking products to enhance the surface view available from ADS-B tracking.
Effect on Airport	ADS-B Out equipment will be available, but is not required, for ground vehicles. If acquired, it can enable all-weather tracking of vehicles in the movement area. To enact an approved surface movement guidance and control system, some airport painting, lighting, signage, or other investments may be required.
Action Needed by Airport	To enable low-visibility operations, the airport must convene a surface movement guidance and control system working group of airport personnel, aircraft operators, and FAA representatives. This group will plan the operating systems and procedures for the airport. (See Advisory Circular 120-57A for a sample and guidance.)
Priority Area	Surface operations and information sharing
More Info	NextGen_Surface_Ops (http://www.faa.gov\nextgen\snapshots\portfolios\?portfolioId=7)

Copyright National Academy of Sciences. All rights reserved.

What	**Improved Approaches and Low-Visibility Operations**
Why	Delays from instrument meteorological conditions, such as fog and marine layer conditions, have impacts across the country. U.S. air travel schedules are based on clear conditions, so delays can quickly become significant. These improvements reduce delays due to bad weather. More efficient descents save fuel and can reduce noise exposure.
How	The GBAS enhances basic GPS signals to a produce 1meter position accuracy and provide an alternative to the ILS, with less stringent siting requirements and lower maintenance. Technologies such as synthetic vision systems (SVSs) and EFVSs in the cockpit and GBASs at the airport enable arrival, departure, and taxi in low-visibility conditions. Required navigation procedures support OPDs and reduce lengthy fuel-inefficient flight at low altitudes during approach.
Who	Aircraft operators with proper equipment and training can serve an airport, including the taxiing process, in low visibility (such as a quarter mile), and some already do so. Aircraft vision equipment used with ILSs has enabled descent to 100 feet altitude before the runway must be acquired with natural vision.
Where	In the near term, ground augmentation is in place at two airports. The SLS-4000 CAT I GBAS system is currently available for purchase and installation by airport authorities as a non-federal navigation aid. The FAA does not plan to procure any systems for airports. The FAA has a CAT III GBAS Research and Development Project underway, but it currently has no plans to develop or field any federal GBAS facilities. Several high-end GA aircraft are equipped with synthetic and enhanced vision systems.
When	Anyone can purchase and install an approved non-federal GBAS CAT I system now. To use it under instrument meteorological conditions (IMC), the owner must follow FAA guidance and evaluation on siting, GLS (GBAS Landing System) procedure development, installation approval, final flight inspection of the station, and each GLS procedure. After final approval, the station would be subject to annual non-federal ground inspections and periodic flight inspections just like any other NAVAID serving in the NAS.
Effect on Airport	One ground augmentation station can support multiple runway ends for ILS-like approaches from several approach paths. This reduces the VHF requirements and simplifies airport infrastructure. The GBAS has more flexible siting criteria, enabling it to serve runways that ILS is unable to support. It requires less frequent flight inspections than ILS. Fewer passengers are stranded in bad weather. SVSs rely on accurate airport geospatial data.
Action Needed by Airport	The FAA does not currently fund GBAS installations. Airports wishing to install GBAS would need to purchase an FAA approved GBAS system and pay for its installation, flight check, and maintenance costs.
Priority Area	Low-visibility operations
More Info	*ACRP Report 150: NextGen for Airports, Volume 1: Understanding the Airport's Role in Performance-Based Navigation: Resource Guide* ACRP Project 03-34 FAA_SatNav (http://www.faa.gov\about\office_org\headquarters_offices\ato\service_units\techops\navservices\gnss\library\satnav\) FAA_GBAS (http://www.faa.gov\about\office_org\headquarters_offices\ato\service_units\techops\navservices\gnss\laas\)

Copyright National Academy of Sciences. All rights reserved.

What **Improved Multiple Runway Operations (MRO)**

1: Closely Spaced Parallel Multiple Runway Procedures

Why This technology allows increased radar arrival capacity on parallel runways.

How Improved wake and blunder analysis for MRO improve the capacity of closely spaced parallel runways by:

(1) Reducing runway centerline separation minimums for simultaneous independent and dependent (staggered) approaches and
(2) Reducing aircraft stagger distance minimums for dependent approaches.

The research covers both dual and triple parallel runways. Some dual runways (3,600 feet apart) can implement simultaneous independent procedures without new equipment. More closely spaced runways (>3,000 feet) need special equipment (such as high-update-rate radar or high-definition displays) to precisely track landing aircraft positions. Dependent approaches for runways <2,500 feet are approved under JO 7110.308A for specific airport and aircraft weight classes. Use of reduced separations requires pilot and air traffic controller training.

Who The FAA is implementing this program through its ATC guidance.

Where Every airport with parallel runways can potentially employ this technology.

When

Operation type	Centerline separations (feet)	Effective date
Dependent arrivals behind large and small category aircraft	<2,500	BOS, CLE, EWR, MEM, SEA, SFO, and STL included on FAA JO 7110.308
Simultaneous independent parallel operations with advanced controller monitoring displays	≥3,600	included in FAA JO 7110.65W
Triple independent parallel operations with advanced controller monitoring displays	≥3,900	

Copyright National Academy of Sciences. All rights reserved.

When	**Operation type**	**Centerline separations (feet)**	**Effective date**
	Dual and triple independent parallel operations with 2.5°-3.0° offset approach for runways with advanced controller monitoring displays	≥3,000	
	Dependent runway stagger distance reduction from 1.5 nm to 1.0 nm	≥2,500 <3,600	
	Dependent runway stagger distance reduction from 2.0 nm to 1.5 nm	>4,300 <7,300	Not yet approved
Effect on Airport	This technology significantly increases capacity in instrument meteorological conditions (IMC) by enabling simultaneous (independent) arrival operations on parallel runways that previously could only use dependent (staggered) arrival operations and the use of dependent parallel operations on closely spaced parallel runways which were previously limited in IMC to single runway operations. In conjunction with high-speed taxiways, this could increase the maximum number of operations at peak periods 1 to 4 percent during the relevant instrument conditions.		
Action Needed by Airport	The amount of the increase depends on airport-specific layout, terrain, and other factors. A feasibility study can reveal the extent to which runway operations could be increased.		
Priority Area	MRO		
More Info	NextGen_MRO_priorities (http://www.faa.gov\nextgen\snapshots\priorities\?area=mro)		

Copyright National Academy of Sciences. All rights reserved.

What	**Improved MRO 2: Wake Turbulence Mitigation**
Why	Wake turbulence mitigation eliminates wake vortex separation requirements imposed by operations on one runway of a parallel pair, which impact operation on the other parallel runway, and increases departure and arrival capacity.
How	Wake turbulence mitigation for arrivals (WTMA) provides that operations on the upwind runways can proceed at minimum diagonal (cross-runway) radar separations instead of observing elongated cross-runway wake-vortex-driven separations. WTMA – Procedures (WTMA-P) allows dependent parallel operations on runways separated by <2,500 feet for specific airports without regard to wind. WTMD allows independent departures on an upwind runway when wind conditions ensure wakes from aircraft on the downwind runway will not affect the upwind parallel runway.
Who	The FAA is implementing this program through its ATC guidance.
Where	WTMA-P is being implemented for parallel runways with <2,500 feet of centerline separation at PHL and DTW, with ATL planned for future. WTMD was authorized for IAH, SFO, and MEM by FAA JO 7110.316, but only SFO appears to be continuing. No implementation information is available for wind-dependent WTMA.
When	As of October 2015, WTMD was determined to be generally not cost effective and was not being pursued. FAA JO 7110.308A, June 1, 2015, authorized WTMA-P at PHL and DTW.
Effect on Airport	Closer spacing may enable 1–10 percent more capacity at no additional cost, depending on the percentage of heavy aircraft traffic.
Action Needed by Airport	These implementations leverage previous airport analyses of layouts and operations. Departure operations require short-term wind productions and use current weather systems. Arrivals require longer-term predictions and may require additional weather equipment. Additional equipment may require electrical and communication cables.
Priority Area	MRO
More Info	https://www.faa.gov/nextgen/snapshots/priorities/?area=mro

Copyright National Academy of Sciences. All rights reserved.

What	**Performance-Based Navigation (PBN)**
Why	PBN allows aircraft to fly more efficient routes through a combination of area navigation, lateral navigation, and vertical navigation capabilities. Area navigation enables use of virtual waypoints. Lateral navigation enables aircraft to conform precisely to the lateral path of a flight procedure. Vertical navigation enables aircraft to (1) conform to altitude restrictions of flight procedures for precise clearance of obstacles and traffic and (2) plan and execute fuel-efficient arrival descents and departure climbs. These and other FAA NextGen initiatives—FIM, trajectory-based operations, and time-based flow management (TBFM)—enable aircraft to fly more efficient routes through a combination of closer separations enabled by self-separation, conflict-free 4D trajectory planning for smooth and direct routes, and taking advantage of time-specific arrival slots in busy airports or busy multi-airport Metroplex airspace.
How	PBN improves capacity and reduces fuel use by enabling more and better optimized departure, en route, and arrival routes due to improved aircraft flight precision. Changes in procedures for RNAV and RNP-equipped aircraft reap these benefits. PBN can also include RNAV or RNP-based curved precision approaches that permit reduced minimums. The FAA can implement these approaches near an airport through the Metroplex program. It can institute Metroplex under a reduced environmental review process, but these approaches have real or perceived noise impacts on the community.
Who	Much planning goes into new satellite-based routes around airports, from FAA flight procedures to ATO mission support. Airport personnel at large and small airports are invited to participate in planning and need to express route concerns through these venues.
Where	PBN Metroplex implementation is complete at Houston. It is being installed at Atlanta, Charlotte, Florida, Northern California, North Texas, Southern California, and Washington, D.C. It is under consideration at Seattle, Las Vegas, Phoenix, Minneapolis, Chicago, Cleveland/Detroit, New York, and Boston.
When	RNAV and RNP have already begun. Establishing new terminal approach routes using RNAV and RNP at major metropolitan areas has been accomplished at North Texas and Houston; other areas are planned for 2015–17.
Effect on Airport	PBN will have potential impacts on capacity, local air quality, noise impacts on the local community, aircraft operator safety, and airport operational costs. Types and degrees of impact vary by implementation, but noise impacts have caused significant public controversy at several locations.
Action Needed by Airport	Large and small airports must stay involved with route planning and be cognizant of the number of routes and use for airport configurations. Some noise-impacted areas will have greatly increased noise. Mitigated noise areas may need to shift, and some people may be exposed to noise for the first time.
Priority Area	PBN

continued

Copyright National Academy of Sciences. All rights reserved.

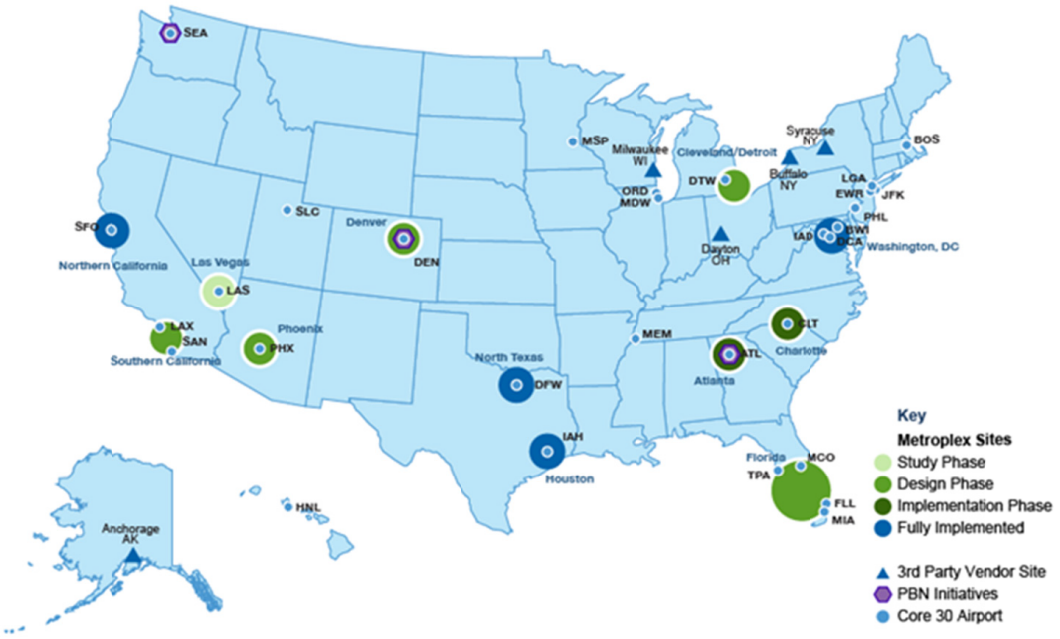
More Info

ACRP Report 150: NextGen for Airports, Volume 1: Understanding the Airport's Role in Performance-Based Navigation: Resource Guide ACRP Project 03-34

FAA_PBN_Progress_and_Plans (http://www.faa.gov/nextgen/update/progress_and_plans/pbn/)

PBN_Status_and_Capabilities (https://www.faa.gov/nextgen/update/operator_investments_and_airports/operator_investments/pbn/)

NextGen_Aircraft_Equipage_Levels (https://www.faa.gov/nextgen/update/operator_investments_and_airports/operator_investments/equipage_levels/)



Metroplex sites (graphic by FAA.)

Copyright National Academy of Sciences. All rights reserved.

What	**Time-Based Flow Management (TBFM)**
Why	Time-based metering can provide smooth flows into the terminal area, minimizing terminal area delays and avoiding lost arrival slot opportunities. Arrival metering can potentially be integrated with departure scheduling to support efficient takeoff and merging into busy overhead airflows. Time-based interval management automation improves arrival capacity by increasing spacing precision and reducing spacing buffers.
How	This program includes upgrades and extensions to the current traffic management advisor (TMA) time-based metering (TBM) system. TBM capability will be enhanced with expanded TMA locations, extended range, incorporation of RNAV/RNP routes and performance, and use of improved data and automation to maximize use of departure slots in congested airspace. Interval management automation enables controllers to safely reduce spacing buffers. This change primarily adds tools to the terminal airspace to complement earlier tools in the en route space.
Who	Northeast and East Coast airports under a busy flow to New York (such as PHL, DTW, MDW, and CLE) have trouble with this now and will experience the benefit. Arrival TBM is anticipated as a far-term improvement.
Where	Installation is limited to computer equipment and displays in air control facilities.
When	This program will take place in the near term (2015–18).
Effect on Airport	This program should streamline arrival and departure flows, increasing capacity to an airport by a small percentage in visual conditions. Interval management tools can potentially support significant improvements in instrument arrival capacity.
Action Needed by Airport	Airports consider this when planning.
Priority Area	PBN
More Info	NextGen_TBFM (https://www.faa.gov/nextgen/snapshots/portfolios/?portfolioId=11) NextGen_Decision_Support_Tools (https://www.faa.gov/nextgen/update/progress_and_plans/decision_support_systems/)

Copyright National Academy of Sciences. All rights reserved.

What	**Collaborative Air Traffic Management (CATM)**
Why	Lack of coordination and data sharing between the FAA and NAS users results in inefficient traffic flow management and avoidable operator costs. Costs include delays and fuel burned when making up time on a longer route.
How	CATM improves NAS operating efficiency and compliance with operator preferences, especially in response to congestion and disruptions, by using up-to-date flight data and collaboration between operator flight planners and FAA traffic managers.
Who	Air traffic controllers, FAA traffic managers, and operators are the intended users. Operators benefit by having their preferences known.
Where	NAS-wide.
When	CATM is currently operational with increasingly better data availability.
Effect on Airport	CATM can improve the service of airport operators.
Action Needed by Airport	The FAA/Industry CDM Stakeholder Surface Team recommended improved data sharing through CDM membership for airports under a revised CDM memorandum of understanding (MOU).[31] Airport data could include construction, snow removal, and, depending on gate ownership agreements, ramp management. Airports should take steps to join the CDM process (telephone calls, meetings, and data sharing), when invited, to enable shared situational awareness.
Priority Area	CDM
More Info	NextGen_CATM (http://www.faa.gov/nextgen/snapshots/portfolios/?portfolioId=12) FAA_CATM_work_packages (http://www.faa.gov/nextgen/programs/catmt/)

[31] Surface Team, "Airport CDM Membership and Improved Data Availability," briefing, RTCA NAC Sub-committee Meeting, August 13, 2014.

Copyright National Academy of Sciences. All rights reserved.

What	**Separation Management 1: Wake Recategorization (Wake RECAT)**
Why	Because of the reduced separation requirements, more aircraft can land and depart on a runway in an hour, increasing capacity.
How	New studies of wake vortex hazards revealed traditional wake hazard separation standards for certain aircraft pairs to be overly conservative, and separations can be safely reduced. Separation reductions focus on Boeing 757 and heavy class aircraft, but some small-behind-heavy separations increase, and benefits depend on the airport traffic mix.
Who	A government-industry working group formulated the recategorization, which will be implemented by FAA through its ATC guidance. Reference JO 7110.659B, Wake Turbulence Recategorization, March 1, 2015.
Where	MEM was the prototype, and SDF, ATL, CLT, CVG, ORD, IAH, CVG, EWR, JFK, and LGA have also implemented it. Implementation will be extended to other airports over time.
When	Limited implementation will began in 2015; consult with the FAA for airport use.
Effect on Airport	Closer spacing may enable 1–10 percent more capacity. MEM achieved a large increase in throughput (19 percent) because it has many lighter heavy class aircraft and could reduce separations on departure. Simulation modeling is used to estimate airport throughput change.
Action Needed by Airport	Airports need to include Wake RECAT in planning analyses, conduct simulation as appropriate, and advocate for recategorization, working with aircraft operators, stakeholders, and the FAA.
Priority Area	Airport enhancement

continued

Copyright National Academy of Sciences. All rights reserved.

More Info

Adapted from Table 5-5-2, JO 7110.659C, Wake Turbulence Separation on Approach

	Follower (nm)					
Leader	A	B	C	D	E	F
A	MRS	5	6	7	7	8
B	MRS	3	4	5	5	7
C	MRS	MRS	MRS	3.5	3.5	6
D	MRS	MRS	MRS	MRS	MRS	4
E	MRS	MRS	MRS	MRS	MRS	MRS
F	MRS	MRS	MRS	MRS	MRS	MRS

Note: MRS = minimum required separation of 2.5 or 3.0 nm.

Category[a]	Capable of an MTOW of (lb.)	Wingspan length (ft.)
A	At least 300,000	Greater than 245
B	At least 300,000	Greater than 175, less than or equal to 245
C	At least 300,000	Greater than 125, less than or equal to 175
D	Less than 300,000	Greater than 125, less than or equal to 175
	Greater than 41,000	Greater than 90, less than or equal to 125
E	Greater than 41,000	Greater than 65, less than or equal to 90
F	Less than 41,000	Less than or equal to 125
	Less than 15,500	Any
	Any powered sailplane	

Note: MTOW = maximum takeoff weight.
[a] Aircraft categories A through F are defined by weight and wingspan dimensions (both dimensions must be true to fit that category).

http://www.faa.gov/documentLibrary/media/Order/JO_7110_659C.pdf
FAA_Wake_RECAT_Fact_Sheet (http://www.faa.gov/news/fact_sheets/news_story.cfm?newsId=18676)

Copyright National Academy of Sciences. All rights reserved.

What	**Separation Management 2: Oceanic In-Trail Procedure**
Why	Because there is no radar over the ocean, aircraft are required to maintain 60 miles of separation from aircraft at a similar altitude over the ocean. Because of the limited information, aircraft have to wait to change altitude until a leader aircraft also changes altitude. ADS-B allows aircraft to see each other and change altitude without waiting for the aircraft in front of them to ascend.
How	The cooperative use of ADS-B between leader and follower aircraft over the ocean allows them to ascend to more efficient flight levels when they can establish no aircraft are nearby to interfere with the new altitude.
Who	Aircraft operators that travel over the ocean are affected. This capability uses a different radio and frequency than en route ADS-B. FANS-1A is the oceanic standard, which has been in place for years.
Where	No additional equipment needs to be installed.
When	This program will take place in the near term.
Effect on Airport	International traffic capacity can increase in small amounts. The primary effect is slightly more efficient fuel use by oceanic flights. More international service is possible as a result of lowering the flight operation cost.
Action Needed by Airport	No action needed.
Priority Area	PBN
More Info	FAA_7110.661_ADS-B_In_trail_Procedure_(ITP) (https://www.faa.gov/regulations_policies/orders_notices/index.cfm/go/document.information/documentID/1025688)

Copyright National Academy of Sciences. All rights reserved.

What	**On-Demand NAS Information**
Why	Critical airport information, such as NOTAMs, are not available electronically at all airports. Digitizing all airport information gives flight crews better and safer flight planning and more seamless taxiing after landing, as part of electronic flight bags.
How	On-demand NAS information supplies digitized data to SWIM, similar to a website for aeronautical information, on the current state of NAS airspace and airports. It includes runway closures, construction, special use airspace closure, temporary availability of special use airspace, and equipment outages. Useful for planning flights, it includes updated airport digital maps and obstacle maps.
Who	Airports are responsible for submitting and updating an airport layout plan to reflect the configuration of the airfield, as well as NOTAMs when operational changes occur. When new flight procedures are designed or a runway configuration changes, airports are also required to submit specific safety-critical data. Increasingly, these data are submitted in an electronic form via the FAA's Airports GIS Program.
Where	This program involves all public airports.
When	Midterm
Effect on Airport	Airports may choose to invest in technologies that enable them to update NOTAMs directly from operations vehicles on the airfield. The method for transmitting and using these data is not yet defined.
Action Needed by Airport	Airports need to obtain an electronic map or plan to gain one through a near-term survey and improvements. They may also invest in additional technologies. The FAA site Airports_GIS (https://airports-gis.faa.gov/public/airportsSteps.html) contains instructions for covering an electronic airport layout plan (eALP) in AIP funding.
Priority Area	Airport enhancement
More Info	NextGen_On-Demand_NAS_Info (http://www.faa.gov/nextgen/snapshots/portfolios/?portfolioId=16)

Copyright National Academy of Sciences. All rights reserved.

What	**Environment and Energy**
Why	The FAA is improving fuel choices for air travel to reduce greenhouse gas emissions and the amount of lead in our environment. The FAA has chosen to bundle many of these programs under NextGen.
How	This program addresses aircraft noise, air quality, climate, energy, and water quality with scientific research, technology and engineering improvements, and environmental management.
Who	Commercial jetliners will increasingly use biofuel. Researchers funded by FAA and NASA are designing technologies to make jet engines more fuel efficient and quieter.
Where	Biofuel refineries are being built around the country. When they reach sufficient capacity, pipeline distributions may need to be constructed, either to connect to existing pipelines or supplement them.
When	Near term. FAA has a goal of production and use of 1 billion gallons of biofuel by 2018. Advanced technology engines are being added to new aircraft as manufacturers offer new aircraft types, so the entire fleet is continuously becoming quieter and more fuel efficient on average.
Effect on Airport	Drop-in fuels are derived from plant matter instead of petroleum reserves. Their reduced environmental footprint comes from recycling greenhouse gases instead of net release. In some cases, drop-in fuels recycle more carbon than they release. Engine technology will reduce the demand for new fuel farms. Low-lead fuels used by GA are targeted for phase-out, and a law phasing out low-lead GA fuel is likely by 2020.
Action Needed by Airport	Drop-in jet fuels are compatible with and designed to mix with commercial jet fuel. No emptying, cleaning, or replacement of existing fuel tank farms is required to use drop-in fuels. Certification testing standards for specific jet fuel blends containing biofuels have been released. Ground vehicles can also run on biofuel. Biofuel use by jets and ground surface equipment will generally not fix a non-attainment zone.
Priority Area	Energy and environment
More Info	NextGen_Environment_&_Energy (http://www.faa.gov/nextgen/snapshots/portfolios/?portfolioId=15) The Voluntary Airport Low Emissions (VALE) program that supports on-airport projects, such as alternate fuel vehicles, at airports in U.S. Environmental Protection Agency non-attainment zones is not part of NextGen. For information on VALE, see FAA_VALE (http://www.faa.gov/airports/environmental/vale/).

Copyright National Academy of Sciences. All rights reserved.

What	**System Safety Management**
Why	To ensure greater safety, FAA is creating prognostics to predict accident precursors. Use of predictive risk tools should allow FAA to target enhancement measures.
How	This program aims to improve safety through data collection and analysis. The primary tool is the Aviation Safety Information and Analysis System (ASIAS).
Who	ASIAS is a data collection and evaluation program conducted by FAA at its headquarters.
Where	Data on the top 50 airports are collected.
When	This program will take place in the near term.
Effect on Airport	Safety data will be collected and made available to interested stakeholders for evaluation and planning.
Action Needed by Airport	No action is needed by airports.
More Info	NextGen_System_Safety_Management (http://www.faa.gov/nextgen/snapshots/portfolios/?portfolioId=17) FAA ASIAS: www.asias.faa.gov/pls/apex/f?p=100:1: (http://www.asias.faa.gov/pls/apex/f?p=100:1) FAA Safety Management System: NextGen_Safety (http://www.faa.gov/nextgen/update/progress_and_plans/safety/)

Copyright National Academy of Sciences. All rights reserved.

What	**NAS Infrastructure**
Why	Bringing NextGen tools to fruition in the field requires study, analysis, planning, drafting of procedures, assessment, and testing. These activities also have to be funded to best implement NextGen technologies.
How	NAS infrastructure provides research, development, and analysis of capabilities that depend on and impact more than one NextGen portfolio. Focus areas include weather and terminal area data production and dissemination, and future ATC technologies.
Who	FAA is assisted by contractors and researchers, who study problems in the airspace, propose a solution, and test its effectiveness before fielding.
Where	All NAS airports are candidates for infrastructure improvements. This work leads to improved systems and data in the en route ATC centers and TRACON facilities.
When	The program will take place in the midterm and far term.
Effect on Airport	Research leads to programs and procedures to respond to weather and disruption, while reducing emissions and flight delay.
Action Needed by Airport	No action is needed by airports.
More Info	NextGen_NAS_Infrastructure (http://www.faa.gov/nextgen/snapshots/portfolios/?portfolioId=14)

Copyright National Academy of Sciences. All rights reserved.

What	**ADS-B In/Cockpit Display of Traffic Information (CDTI)**
Why	ADS-B In is a capability that provides safety benefits on its own and enables airborne separation management capabilities.
How	ADS-B In consists of an ADS-B Mode S or UAT receiver and a CDTI. The CDTI displays ADS-B traffic information directly from other ADS-B–equipped traffic and radar data from non-ADS-B transponder-equipped aircraft transmitted by the ground-based FAA traffic information system–broadcast (TIS-B). The ground-based traffic information system–rebroadcast (TIS-R) broadcasts data from Mode A, C, and S transponder-equipped aircraft to UAT equipped aircraft, and vice versa. CDTI data can be displayed on a multifunction display with a moving map and weather data from the ground-based FAA flight information system–broadcast (FIS-B). ADS-B In enhances safety by giving pilots visibility of traffic in the air and with ADS-B–equipped surface vehicles on the ground. Regarding efficiency, pilots with CDTI can follow traffic on approach through intermittent losses in visibility, thus extending the use of visual approaches and, with additional automation, pilots can perform precision airborne merging and spacing with other traffic.
Who	Aircraft with Mode S or UAS ADS-B equipment aircraft can add CDTI. FIS-B data are only broadcast to UAT transceivers.
Where	ADS-B In can provide traffic information from ADS-B–equipped aircraft using the same radios (Mode S or UAT) without ground equipment.
When	ADS-B In is optional; there are no mandates for ADS-B In.
Effect on Airport	ADS-B In improves safety both on the surface and in the terminal airspace by providing traffic and weather information to pilots.
Action Needed by Airport	No action is needed by airports. Equipage of surface vehicles with ADS-B Out to enhance surface movement safety is an optional step.
More Info	*ACRP Report 150: NextGen for Airports, Volume 1: Understanding the Airport's Role in Performance-Based Navigation: Resource Guide* (ACRP Project 03-34) NextGen_ADS-B (http://www.faa.gov/nextgen/update/operator_investments_and_airports/operator_investments/adsb/)

Copyright National Academy of Sciences. All rights reserved.

What	**Wide-Area Augmentation System (WAAS)**
Why	The basic GPS/GNSS navigation accuracy is not good enough to support precision approaches.
How	WAAS provides corrections to GPS/GNSS data that increase accuracy from 20 meters to 1.5–2 meters, which is sufficient to enable approaches with ILS CAT I minimums. Ground stations throughout the U.S. record GPS/GNSS signals and transmit them to a central facility that processes correction data. The correction data are transmitted to satellites that broadcast them to aircraft on the GPS L1 frequency.
Who	Aircraft with WAAS-augmented GPS/GNSS can use LPV guidance approaches that have minimums equivalent to ILS CAT I, without an installed ILS localizer or glide slope.
Where	WAAS covers the entire continental U.S. including Alaska.
When	As of May 26, 2016, there were 3,678 LPV approaches, serving 1,790 airports, including 1,041 non-ILS airports.[32] There were more than 77,500 GA aircraft equipped with WAAS.[33]
Effect on Airport	WAAS-enabled ILS CAT I equivalent approaches significantly increase arrival capacity at non-ILS airports and runways.
Action Needed by Airport	No action is needed by airports. However, if a new GPS approach reduces minimums it may trigger more stringent obstacle clearance requirements, which may need to be undertaken by the airport.
Priority Area	PBN
More Info	*ACRP Report 150: NextGen for Airports, Volume 4: Leveraging NextGen Spatial Data for Airports: Guidebook* (ACRP Project 09-12) *ACRP Report 150: NextGen for Airports, Volume 1: Understanding the Airport's Role in Performance-Based Navigation: Resource Guide* (ACRP Project 03-34) *SatNav*: GNSS_Library_SatNav_News (http://www.faa.gov/about/office_org/headquarters_offices/ato/service_units/techops/navservices/gnss/library/satnav/) *WAAS*: FAA_WAAS (http://www.faa.gov/about/office_org/headquarters_offices/ato/service_units/techops/navservices/gnss/waas/)

[32]FAA Satellite Navigation—GPS/WAAS Approaches, www.faa.gov/about/office_org/headquarters_offices/ato/service_units/techops/navservices/gnss/approaches/.

[33]NextGen Performance-Based Navigation, www.faa.gov/nextgen/update/progress_and_plans/pbn/.

Copyright National Academy of Sciences. All rights reserved.

What	**Ground-Based Augmentation System (GBAS)**
Why	WAAS is not accurate enough the support ILS CAT II/III equivalent approaches.
How	GBAS provides corrections to GPS/GNSS data that increase accuracy. Current GBAS systems allow ILS CAT I equivalent approaches, but the research goal is to enable ILS CAT III approaches. GBAS uses three or more ground stations in the vicinity of the airport to record GPS/GNSS signals and transmit them to a central facility that processes correction data. The correction data are transmitted to aircraft by VHF radio. A single VHF can support up to 48 procedures on different runways.[34] The locally generated corrections are more accurate than the broadly based WAAS corrections.
Who	Aircraft with GBAS-augmented GPS/GNSS can use GLS approaches without an installed ILS localizer or glide slope.
Where	GBAS will be most useful for airports needing ILS CAT III capability that cannot support an ILS. GBAS is also useful in areas not covered by WAAS, such as the Caribbean.
When	GBAS ground equipment and avionics are commercially available. Current GBAS Approach Service Type–Level C (GAST-C) supports ILS CAT I equivalent approaches, and ICAO GAST-D standards supporting CAT III equivalent approaches are expected to be finalized by 2019.[35]
Effect on Airport	GBAS-enabled ILS CAT III equivalent approaches will significantly increase arrival capacity at non-ILS airports and runways.
Action Needed by Airport	Currently, all GBAS systems are non-federal, purchased by airport authorities or other private parties. The purchasers must coordinate receiver locations and other implementation details with the FAA to qualify for use of GAST approaches.[36]
Priority Area	PBN
More Info	SatNav: GNSS_Library_SatNav_News (http://www.faa.gov/about/office_org/headquarters_offices/ato/service_units/techops/navservices/gnss/library/satnav/) GBAS: FAA_GBAS (http://www.faa.gov/about/office_org/headquarters_offices/ato/service_units/techops/navservices/gnss/laas/)

[34]FAA, GNSS Frequently Asked Questions—GBAS, www.faa.gov/about/office_org/headquarters_offices/ato/service_units/techops/navservices/gnss/faq/laas/.
[35]FAA, GNSS Frequently Asked Questions.
[36]FAA, GNSS Frequently Asked Questions.

Copyright National Academy of Sciences. All rights reserved.

What	**Wide-Area Multilateration (WAM)**
Why	WAM provides radar-like surveillance of transponder-equipped aircraft where radar coverage is not available, such as in mountainous terrain.
How	WAM uses multiple sensors to interrogate aircraft transponders and send the timed returns to a central processor to triangulate the position of the aircraft. In the future, ADS-B will provide similar coverage to WAM, and WAM can serve as a back-up to ADS-B.
Who	All aircraft with transponders can be tracked by WAM.
Where	WAM is beneficial for areas where terrain prohibits the use of radar.
When	WAM systems are currently in use at Juneau, AK, and several airports in the Colorado Rocky Mountains.[37]
Effect on Airport	WAM in Colorado has enabled 12–17 operations per hour in bad weather compared with 4 without it. At Juneau, WAM enables use of 5-mile radar-equivalent separations through the Gastineau Channel.
Action Needed by Airport	Local purchase may be required. The Rocky Mountain WAM system was purchased by the State of Colorado and turned over to the FAA.[38]
Priority Area	PBN
More Info	*ACRP Report 150: NextGen for Airports, Volume 1: Understanding the Airport's Role in Performance-Based Navigation: Resource Guide* (ACRP Project 03-34) FAA_WAM

[37]FAA WAM Project, www.faa.gov/nextgen/programs/adsb/wsa/wam/.
[38]Ibid.

Copyright National Academy of Sciences. All rights reserved.

What	**Flight-Deck Interval Management (FIM)**
Why	Improved spacing precision and reduced controller workload will be necessary to handle increased future demands and PBN trajectory management.
How	FIM uses ADS-B In data from leader aircraft to accomplish precision in-trail and merge point spacing. Aircraft automation calculates speed commands to maintain controller-assigned in-trail and merging time intervals. Precision is increased due to the accuracy and low latency of ADS-B data, and controller workload is reduced because the aircraft are controlling spacing without controller speed or vectoring commands.
Who	FIM equipped aircraft can establish spacing on ADS-B–equipped leader aircraft.
Where	FIM operations would be available both en route and in the terminal area. FIM has been flight tested by NASA and UPS has used FIM procedures experimentally at Louisville.
When	NASA and FAA research is ongoing to develop procedures for in-trail spacing, merging, and paired approaches to closely spaced parallel runways, but no implementation dates are set.
Effect on Airport	FIM operations could substantially increase runway and airport arrival capacity in instrument conditions.
Action Needed by Airport	No action is needed by airports.
More Info	FAA_ADS-B (http://www.faa.gov/nextgen/update/operator_investments_and_airports/operator_investments/adsb/)

Copyright National Academy of Sciences. All rights reserved.

What	**Surface Management**
Why	Queuing aircraft on taxiways to await runway departures is inefficient, wastes fuel, creates emissions, and frustrates passengers. Departure metering keeps aircraft at the gate until they can taxi unimpeded to the runway for takeoff.
How	Administered at the airport during runway or taxiway closures, this program uses historical data under various weather conditions to predict departure demand and determines 15-minute departure slots, 2 hours in advance. It then assigns these departure slots to aircraft operators on the basis of their scheduled number of departures. In some instances, the aircraft operator is free to assign any of its company's flights to the departure slot. This gives the aircraft operator, ramp towers, or airport operator the tools to make key decisions on the most efficient movement of the airplanes.
Who	JFK provided a prototype operation. SFO and EWR are using the system to diminish queuing and congestion during runway reconstruction, on a trial basis. Variants of the concept, in which aircraft operators had more decision space about which flight would taxi, were also tested at BOS and MCO. A test is planned at LGA.
Where	The JFK system uses ASDE-X radar, an FAA data feed, and commercial software tools in an airport operations center.
When	JFK has used departure metering since 2010. SFO and EWR started in summer 2014.
Effect on Airport	The primary benefit is reduced time in departure queues and reduced emissions. Shorter queues may also provide more flexibility for dealing with taxi disruptions and configuration changes.
Action Needed by Airport	If gate space is over capacity, additional parking stands may be needed to "hold" aircraft that under today's procedures would have pushed back and waited on the ramp and apron. The airport operators may also be directly involved in ramp management and need to add personnel and software to support coordination.
Priority Area	Surface operations and information sharing
More Info	FAA JFK: NextGen_Surface_Management_at_JFK (http://www.faa.gov/nextgen/snapshots/stories/?slide=2)

Copyright National Academy of Sciences. All rights reserved.

What	**Airport Geographical Information System (GIS)**
Why	Accurate geophysical (aka, geospatial) data are essential for safe surface and airborne operations. Both safety-critical and non-safety-critical data are being collected in electronic databases for both current and future airport operations including those using NextGen technologies.
How	"Geospatial data consists of airport features such as navigational aids, taxiways, and aprons as well as potential obstacles and features of landmark value used for general orientation, including shorelines, roads, and railroads. The collection of the features must adhere to these topological and cartographic rules to ensure quality. These geospatial data features, when entered into the FAA Airport Surveying–GIS Program database, provide a foundation for GIS analysis, approach procedure design, and content to create electronic Airport Layout Plans (eALPs) and aeronautical charts."[39]
Who	The FAA Office of Airport Safety and Standards (AAS-1) administers the program. Airports are responsible for airport GIS development and updating.
Where	This program involves all public airports.
When	Current
Effect on Airport	The FAA memo, "Airports GIS Transition Policy for Non-Safety Critical Projects," August 23, 2012, provides guidance for the provision of safety-critical data for certain new undertakings into the GIS for all airports in the NPIAS and phased incorporation of non-safety-critical data for specific airport classes in the NPIAS. Cost is recognized as a major issue, and is discussed in the accompanying ACRP Project 09-12 report: *ACRP Report 150: NextGen for Airports, Volume 4: Leveraging NextGen Spatial Data for Airports: Guidebook.*
Action Needed by Airport	Airports should develop mapping data in accordance with the requirements of the FAA's Airports GIS Program. The Airport_GIS web links below provide information on GIS, eALP, and funding.
More Info	*ACRP Report 150: NextGen for Airports, Volume 4: Leveraging NextGen Spatial Data for Airports: Guidebook* (ACRP Project 09-12) FAA_GIS_policy_memo (http://www.faa.gov/airports/planning_capacity/airports_gis_electronic_alp/media/airportsGISTransitionPolicy.pdf) http://www.faa.gov/airports/aip/aip_handbook/ FAA_Implementation_Guidance_for_Airports_GIS_and_eALP (http://www.faa.gov/airports/planning_capacity/airports_gis_electronic_alp/) FAA_GIS_Advisory_Circulars (http://www.faa.gov/airports/resources/advisory_circulars/index.cfm/go/document.list/parentTopicID/187)FAA_Airports_GIS_Survey (https://airports-gis.faa.gov/public/airportsSteps.html) FAA_Airports_GIS_Public (https://airports-gis.faa.gov/public/airportsSteps.html) FAA_GIS_Info (http://airports-gis.info/) NPIAS_airports (http://www.faa.gov/airports/planning_capacity/npias/) Part_139_airports (http://www.faa.gov/airports/airport_safety/part139_cert/)

[39]FAA AC 150/5300-18B, Dated 5/21/2009, Section 3.1, pg 67.

Copyright National Academy of Sciences. All rights reserved.

4 NextGen Airport by Airport

General Considerations

NextGen benefits will apply differently to different airports. The applicability of NextGen technologies for airports is affected by such characteristics as type of traffic, existing navigation aids, and terrain. For example,

- Airports with high traffic demand can benefit from improved surface operations and data sharing and surface collaborative decision making (Surface CDM).
- Airports with all levels of traffic can benefit from the ability of Data Comm to support higher departure rates and the ability to recover from convective weather events.
- Airports with high traffic demand that includes larger fractions of Heavy and B757 Class aircraft can benefit from the increased capacity provided by Wake RECAT.
- Airports with high traffic demand and constrained departure paths can benefit from additional equivalent lateral spacing operation (ELSO)-enabled departure paths.
- Airports with parallel runways having centerline separations of less than 4,300 feet can benefit from increased capacity in instrument conditions.
- Airports without ILS due to economics or geographic limitations can benefit from satellite-based instrument approaches.

The potential use of NextGen procedures at an airport is also dependent on the equipage of the aircraft using the airport. Table 4-1 shows the equipage levels reported on the FAA PBN website as of June 2016.

Table 4-1. FAA PBN equipage levels.

CURRENT EQUIPAGE LEVELS OF AVAILABLE ENABLERS			
Enabler	Air Transport	Air Taxi	Helicopter
RNP 4	45%***	49%***	AAR**
RNAV 1	97%	98%	67%
RNAV 2	98%	98%	67%

continued

Copyright National Academy of Sciences. All rights reserved.

Table 4-1. Continued

CURRENT EQUIPAGE LEVELS OF AVAILABLE ENABLERS			
Enabler	Air Transport	Air Taxi	Helicopter
RNP Approach	88%	98%	67%
RNP 1 with Curved Path	59%	8%	<1%
VNAV	87%	30%	1%
LPV Approach	AAR**	AAR**	AAR**
RNP AR Approach	62%	5%	<1%
ADS-B Out (rule compliant)	5%	8%	5%
Airborne/Ground CDTI, ADS-B In	1%	4%	2%
ITP, ADS-B In	1%***	4%***	AAR**
FANS 1/A (SATCOM)	32%***	14%***	AAR**
FANS 1/A (VDL Mode 2)	12%	2%	<1%
HUD/ILS	23%	N/A*	N/A*
EFVS	1%	N/A*	N/A*
EFB	88%	N/A*	N/A*

*N/A means flight plan data for specific capability is unavailable.
**AAR indicates additional analysis is required.
***This indicates oceanic capability and percentage represents equipage for aircraft seen operating over oceanic airspace.
Source: https://www.faa.gov/nextgen/update/operator_investments_and_airports/operator_investments/equipage_levels/.

Airports Near a Major Hub with Overlapping Approaches

As discussed in Chapter 1, PBN RNAV/RNP can be used to develop additional arrival and departure waypoints and flight corridors, in order to de-conflict traffic to adjacent airports and increase capacity. This has been experienced at DFW and DAL, JFK and LGA, and ORD and MDW. The additional flight corridors enabled by ELSO have increased departure capacity at ATL. Major hubs and Metroplex airports are used primarily by commercial aviation and aircraft can be expected to be equipped, qualified, and motivated to use the PBN procedures, and all commercial flights are required to file instrument flight plans and follow instrument flight procedures.

Copyright National Academy of Sciences. All rights reserved.

Airports with One or No ILS

NextGen technologies can provide a range of low-visibility approaches on all runways, without the installation and recurring calibration expense of an ILS. Lights and markings may still be required. Curved RNP routes can be designed to avoid terrain obstacles and provide instrument approaches for runways that do not have sufficient straight line obstacle clearance for ILS installation.

The GA aircraft that typically use the smaller airports are not as likely as commercial aircraft to be equipped to use the NextGen technologies; however, there is a trend toward WAAS equipage in the GA fleet and the ADS-B mandate will also likely increase the fraction of adequately equipped GA aircraft

- LPV minimums, usually 200 or 250 feet above ground level, are typically the lowest available on a GPS approach. Other less demanding technologies have higher minimums and may only provide lateral guidance, but still provide improvements over existing procedures
- GBAS is currently available for CAT I equivalent approaches; higher categories are being studied for future availability. GBAS potentially can provide ILS CAT II/III equivalent approaches at all runways of an airport. Lights and markings will still be required in accordance with FAA Order 8900.[40]

LNAV minimums are used to fly a series of stair steps until you reach a minimum descent altitude. They are the only straight-in GPS minimums that can be flown with a non-WAAS, approach-certified GPS. Circling minimums permit you to circle to land on any runway, except those prohibited in the chart notes.

Note that as the FAA digitizes and updates its own digital obstacle database, particularly if an airport applies for a new approach, the FAA flight procedures office may require the airport to clear an obstacle on the approach path or mitigate its risk to the approach (e.g., displaced threshold.). AIP funding can be used to fund a survey of a known or suspected obstacle to determine whether it impinges the flight path.

Airports with Multiple ILS

Commercial airports that currently have multiple obstacle-free, Category II/III ILS may not be interested in the curved approach benefits of RNAV/RNP and the potential low ceiling and visibility capability of GBAS because they have adequate all-weather capability, but they may find GBAS-based approaches (GLS) useful back-ups to cover ILS outages. The RNAV/RNP approaches are also designed with seamless connections to the RNAV STARs.

Airports with Predominantly GA Traffic

GA traffic, like all traffic in the specified controlled airspace, falls under the mandate to equip with ADS-B Out. Aircraft that fly under 18,000 feet have the choice of acquiring UATs that operate at 978

[40]FAA Order 8900.1, Flight Standards Information Management System, Volume 4, Chapter 2, Section 7, 4-337B.

Copyright National Academy of Sciences. All rights reserved.

MHz or a Mode S that operates at 1,090 MHz. On January 1, 2020, aircraft without ADS-B Out equipage will not have access to most controlled airspace.

Aircraft that also install the receiving capability (ADS-B In) can receive traffic awareness from TIS-B and weather information maps from FIS-B. TIS-B and FIS-B services are available across most of the U.S. The ADS-B In avionics needed to provide TIS-B traffic awareness capability cost less than other traffic awareness systems available. In addition, the use of FIS-B weather data involves no monthly subscription fee.

For some GA operators, the ability to receive free traffic and weather information is a welcome change. Mixed equipage, in which some aircraft are equipped and some are not, is likely for the foreseeable future. This may be a topic of discussion at non-towered airports, with one set of pilots self-separating with heavy reliance on their traffic display, and another set self-separating using traditional methods.

Airports Without Radar Surveillance

These airports benefit from increased capacity due to improved surveillance. Without radar, WAM, or ADS-B surveillance, IFR traffic operates as one-on, one-off. WAM and ADS-B enable capacities close to radar-IFR levels.

Installation of WAM allows controllers to guide transponder-equipped (ATC Radar Beacon System Mode C transponders or better) aircraft safely where low-lying radar coverage does not exist. Using triangulation, ATC can locate and separate aircraft, enabling approach in low ceilings, particularly in mountainous terrain where lack of radar and weather previously required greater spacing and caused long delays. WAM must be purchased by the airport, state, or local aviation authority, for integration to FAA ATC.

Airports Planning Improvements

In accordance with FAA guidance mentioned in previous chapters, as of FY 2015, Part 139 airports with ATC towers are required to provide GIS data for all safety-critical and non-safety-critical changes.[41] Some airports with a high number of operations will need to file GIS information with their master plan update. Most airports will need only to file GIS information relevant to the improvement being planned, such as GIS information on the runway end being improved. GIS information in this case includes exact location of the runway end, runway threshold, centerline, and location of any obstacles on the approach path.

Improved GIS information is provided to enable certain GPS-based procedures such as RNP approaches and departures, situational awareness, digital NOTAMs for the airport digital flight bag information, and updated maps to avionics when the aircraft supports it. For example, more exact information about runway thresholds and ends allows ATO to approve tighter spacing of aircraft on approach or departure, particularly in MRO.

The ACRP NextGen initiative reports listed below contain information specific to the array of airports in the NAS.

[41] FAA_GIS_policy_memo, and Part_139_airports.

Copyright National Academy of Sciences. All rights reserved.

ACRP Report 150: NextGen for Airports, Volume 1: Understanding the Airport's Role in Performance-Based Navigation: Resource Guide (ACRP Project 03-34)

ACRP Report 150: NextGen for Airports, Volume 2: Engaging Airport Stakeholders: Guidebook (ACRP Project 01-28)

ACRP Report 150: NextGen for Airports, Volume 4: Leveraging NextGen Spatial Data for Airports: Guidebook (ACRP Project 09-12)

ACRP Report 150: NextGen for Airports, Volume 5: Airport Planning and Development (ACRP Project 03-33)

Table 4-2 lists the airports and agencies surveyed as case studies in the *ACRP Report 150: NextGen for Airports* series, along with the type of information addressed in the survey.

Table 4-2. Case study survey airports.

AIRPORT CODE	AIRPORT NAME	GIS	PBN	PLANNING	STAKEHOLDERS
ATL	Atlanta Hartsfield, GA		Departure		
BFI	Boeing Field, WA				NextGen Plans
DEN	Denver International, CO		Departure, Arrival, Approach		Metroplex
DFW	Dallas-Ft. Worth, TX	X			RNAV
EWR	Newark, NJ			GBAS	
HEN	Henderson, NV		Departure, Arrival, Approach		
IAH	Houston International, TX		Metroplex, Departure, Arrival, Approach		
JFK	New York Kennedy, NY			CDM	
LGA	New York LaGuardia, NY				TNNIS Departure
BVY	Beverly, MA				RNAV, STARS, LPV
MSP	Minneapolis-St. Paul, MN		Departure, Arrival		
PAE	Paine Field, WA				PBN
PDX	Portland, OR	X			
ORD	Chicago O'Hare, IL				New Runways
SEA	Seattle-Tacoma, WA	X	Departure, Arrival, Approach		PBN

continued

Copyright National Academy of Sciences. All rights reserved.

Table 4-2. Continued

AIRPORT CODE	AIRPORT NAME	GIS	PBN	PLANNING	STAKEHOLDERS
SFO	San Francisco International, CA			SOIA	
SUN	Friedman Memorial, ID			RNP Approach	
	CDOT, CO			WAM	
	Puget Sound, WA			NextGen	
	Eurocontrol				Collaboration
	TXDOT, TX	X			
	Washington State, WA	X			

Copyright National Academy of Sciences. All rights reserved.

5 Working with the FAA

NextGen evolves from year to year. For example, sometimes runway closures and rehabilitations drive a new implementation, such as deployment of departure taxi metering at SFO. Further, implementation dates change and the priority of programs and locations is rearranged.

This guide briefly explains NextGen, its history, its intent, and how it works, and links to information sources, but it does not capture everything. The *ACRP Report 150: NextGen for Airports* series reports provide more in-depth information on working with the FAA. Table 5-1 lists points of contact for obtaining more information on NextGen, specifically the Metroplex programs, which have impacted airports.

Table 5-1. Points of contact for more information.

TO FIND OUT ABOUT ...	CONSULT ...
Exact routes and charting for routes for a Metroplex	FAA-ATO Flight Procedure Teams
Planning level activity for routes for a Metroplex	FAA-ATO Office of Mission Support
Schedule of implementation projects other than flight procedures	FAA-ATO Program Management Office
GPS-enabled LPV approaches	FAA-FPO National Flight Procedures Group Instrument Flight Procedures Information Gateway https://www.faa.gov/air_traffic/flight_info/aeronav/procedures/
Near-term implementation plans (where, when) for NextGen	www.faa.gov/nextgen (http://www.faa.gov/nextgen)
Metroplex/PBN routing implementation schedules	www.faa.gov/nextgen/library
More information about particular NextGen programs, such as PBN	FAA's NextGen Snapshots, www.faa.gov/NextGen/snapshots/ (http://www.faa.gov/NextGen/snapshots/) and www.faa.gov/NextGen/programs/ (http://www.faa.gov/NextGen/programs/)

continued

Copyright National Academy of Sciences. All rights reserved.

Table 5-1. Continued

TO FIND OUT ABOUT …	CONSULT …
FAA funding mechanisms for airport improvements related to NextGen programs	FAA-APP, FAA Office of Airports
Public-private partnership for aircraft equipage	www.nexacapital.com (http://www.nexacapital.com/)
The latest comprehensive list of NextGen programs in near, mid, and far terms	nasea.faa.gov
NAC findings	www.rtca.org/
Electronic airport maps: what is required, and a sample statement of work to hire a contractor to perform the work	https://airports-gis.faa.gov/public/airportsSteps.html

Note: ATO = Air Traffic Organization; FPO = Flight Procedures Office.

Figure 5-1 shows the FAA's published schedule for Metroplex implementation (from the 2015 NextGen Implementation Plan). Note that plans are for Las Vegas to be added to the list.

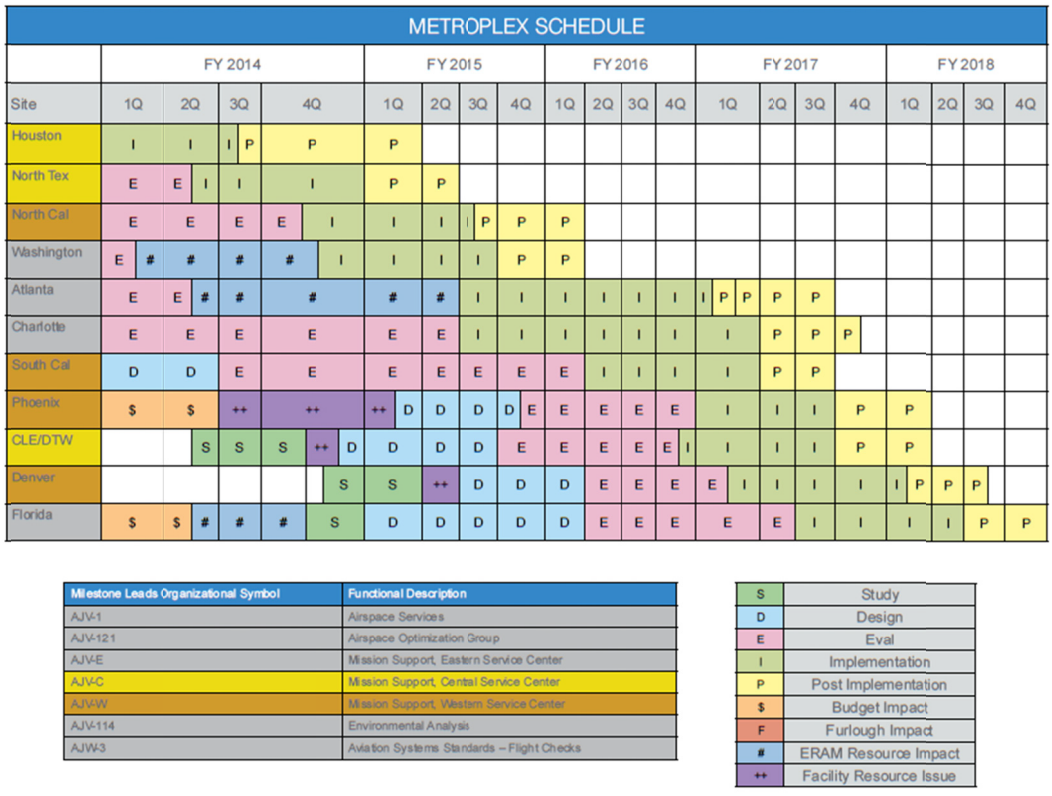


Figure 5-1. *Metroplex schedule chart.*

Copyright National Academy of Sciences. All rights reserved.

Where NextGen EAs Originate in the FAA

The use of EAs, environmental impact studies, and noise studies for airports undergoing changes in infrastructure or flight procedures varies depending on the reason for the change. Figure 5-2 shows the origination of NextGen program actions in the FAA. The origination point of various actions determines the level of environmental studies and analysis the FAA will undertake as NextGen initiatives roll out.

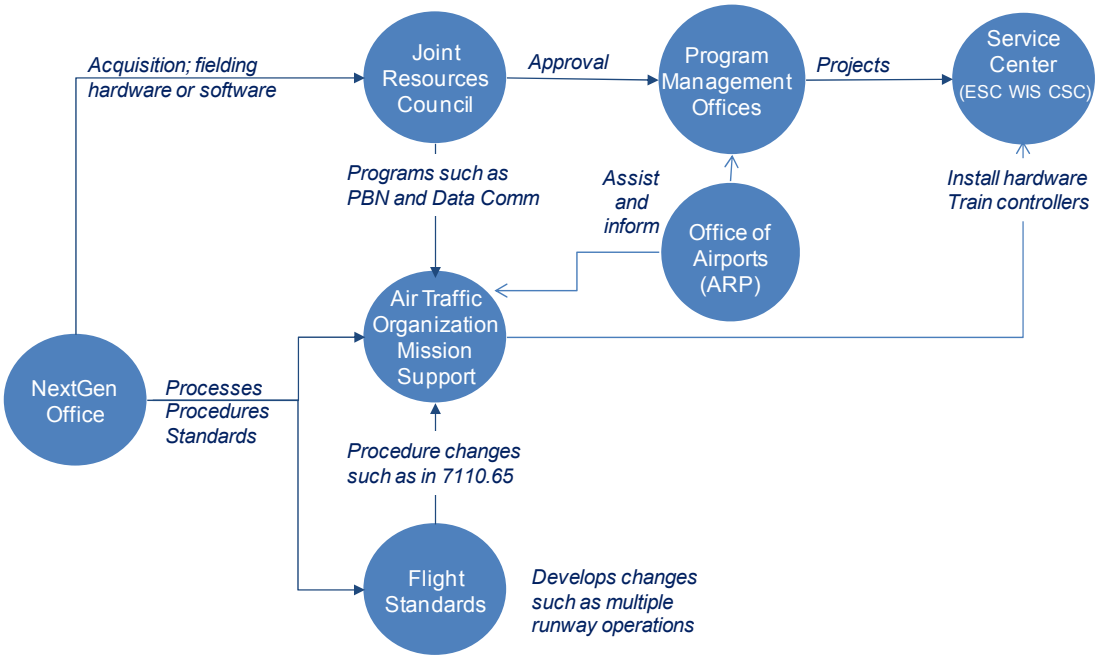


Figure 5-2. *NextGen implementation workflows in FAA offices.*

NEPA (National Environmental Policy Act) Process Impacts

The office that initiates an action determines how the environmental processes are conducted in compliance with that organization's environmental order, specifically addressing the level of environmental analysis and the extent of outreach. A NEPA process is required for federal actions. Figure 5-3 shows the process the FAA uses when it leads a NEPA effort. Under project initiation, FAA begins the NEPA process when it learns of any contract, lease, construction, rulemaking, certification, regulatory action, licensing, permits, or plans submitted for approval. The FAA begins a NEPA evaluation for an airport when it becomes involved, such as when an airport submits an airport layout plan or an AIP grant application for FAA approval.

Some actions, which have minimal impact, will quality for CatEx from EA. There are a number of these exclusions. The more straightforward exclusions include such actions as repairing an existing facility. More involved exclusion decisions include such actions as evaluating new routes for one airport in a Metroplex when the impacts have already been studied under another airport. An environmental

Copyright National Academy of Sciences. All rights reserved.

impact statement can be started right away if the FAA believes the environmental impacts could be significant. As discussed in the *ACRP Report 150: NextGen for Airports, A Primer*, while PBN programs are eligible for CatExs, the March 29, 2016, FAA policy memo states that, "Collaboration with airport operators and public notification should include provision for appropriate community outreach that not only informs the affected public of the FAA's proposal, but also allows the public to provide feedback on community concerns."[42]

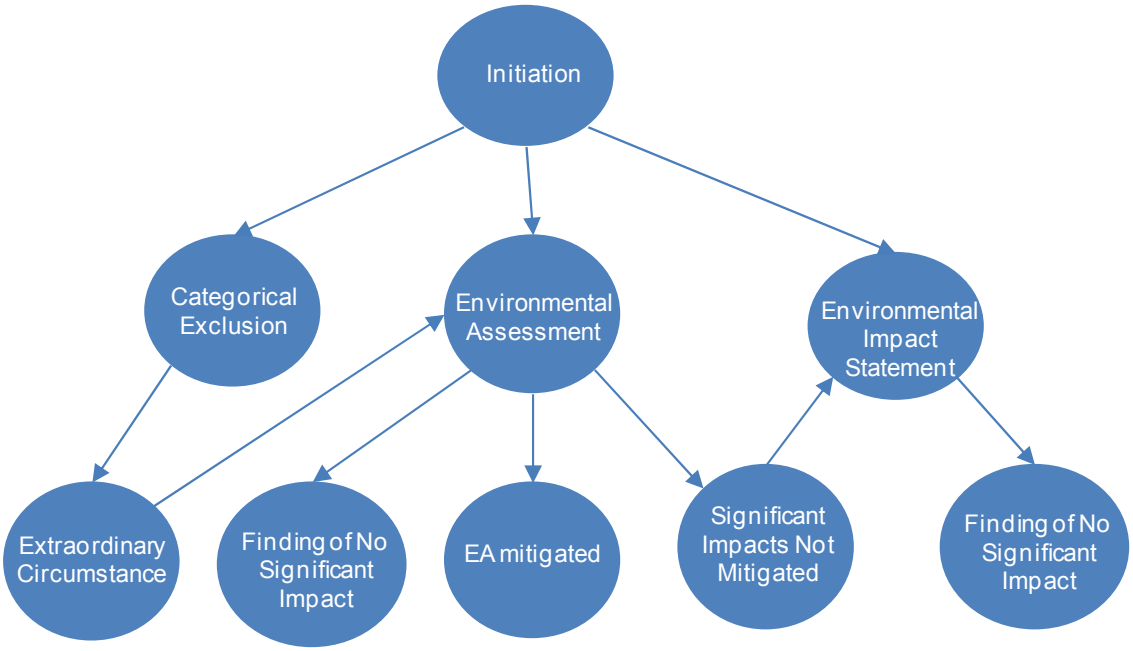


Figure 5-3. *Initiation of the NEPA process.*

The process for FAA-initiated plans is the same as that for airport-initiated actions, but the oversight changes depending on the project owner as does the discretionary public outreach.

The FAA Office of Airports leads EAs for airport actions. The Office of Environment and Energy (AEE) is the NEPA liaison in change of supplying noise and air quality modeling tools. However, ATO leads EAs for Air Traffic Operations, including many PBN implementations. There are important differences in EA practice.

An airport typically holds meetings, sends newsletters, notifies public officials, initiates focus groups, and often chooses to hold a public hearing for significant airport changes. NEPA guidance requires ATO to hold a public hearing at the "earliest" stages of project planning. Historically, ATO advertises the meeting in the public newspaper as required by NEPA but does not have to pursue additional methods of notification; however, ATO may be changing this policy based on industry recommendations. Another key factor is that by holding a public hearing early in the process of, for example, implementing proposed airspace routes, the public may not perceive any changes as imminent or important, until they actually occur. In contrast, typically an airport will choose to engage its community over and over again to collect ongoing commentary on the project.

The result is that an early public notification of changed air routes does not as effectively involve the community and mitigate their concerns as does a repeated outreach process conducted according to the Office of Airports environmental order. Members of the community may have failed to see, notice,

[42]FAA Order 1050.1, Guidance Memo (New CATEX2 Guidance), March 29, 2016, located at https://www.faa.gov/about/office_org/headquarters_offices/apl/environ_policy_guidance/policy/faa_nepa_order/.

Copyright National Academy of Sciences. All rights reserved.

or attend the required public hearing under the ATO process and are surprised when air routes and aircraft noise change.

> Refer to *ACRP Report 150: NextGen for Airports, Volume 2: Engaging Airport Stakeholders: Guidebook* (ACRP Project 01-28) for more information on community outreach.

Summary

The FAA is primarily entrusted with ensuring safety in national airspace. It faces competing pressures from aircraft operators to accomplish airspace and flight procedure changes quickly as well as the legal requirements of a diligent process. Airports can assist in NextGen implementation in their communities by being involved and providing extra communication links for the public about these ongoing changes.

Copyright National Academy of Sciences. All rights reserved.

Copyright National Academy of Sciences. All rights reserved.

A ACRONYMS AND ABBREVIATIONS

ACRONYM	DEFINITION
ACARS	Aircraft Communication Addressing and Reporting System
ADF	Automatic Direction Finder
ADS-B	Automatic Dependent Surveillance–Broadcast
AEE	Office Of Environment And Energy
AEFS	Advanced Electronic Flight Strips
AFS	AVS Flight Standards
AFSS	Automated Flight Service Stations
AIP	Airport Improvement Program
AR	Authorization Required
ARTCC	Air Route Traffic Control Center
ASDE	Airport Surface Detection Equipment
ASDE-X	Airport Surface Detection Equipment-Model X
ASIAS	Aviation Safety Information and Analysis System
ASSC	Airport Surface Surveillance Capability
ATC	Air Traffic Control
ATM	Air Traffic Management
ATO	Air Traffic Operations
ATS	Air Transportation System
AVS	Aviation Safety
CAP	Civil Air Patrol
CatEx	Categorical Exclusion
CATM	Collaborative Air Traffic Management
CDM	Collaborative Decision Making
CDTI	Cockpit Display of Traffic Information
CNS	Communication, Surveillance, and Navigation Technologies
ConOps	Concept of Operations
CPDLC	Controller-Pilot Data Link Communications
CSG	CDM Stakeholders Group
CSPR	Closely Spaced Parallel Runway
DCL	Departure Clearance
DME	Distance Measuring Equipment

Copyright National Academy of Sciences. All rights reserved.

ACRONYM	DEFINITION
DSR	Display System Replacement
EA	Environmental Assessment
eALP	Electronic Airport Layout Plan
EFVS	Enhanced Flight Vision Systems
EoR	Established on RNP
ERAM	En Route Automation Modernization
FAA	Federal Aviation Administration
FAF	Final Approach Fix
FANS	Future Air Navigation Systems
FBO	Fixed-Base Operator
FIDS	Flight Information Display System
FIM	Flight-Deck Interval Management
FIS-B	Flight Information Service Broadcast
FPO	Flight Procedures Office
FTI	Flight Telecommunications Infrastructure
GA	General Aviation
GAST-C	GBAS Approach Service Type–C
GAST-D	GBAS Approach Service Type–D
GBAS	Ground-Based Augmentation System
GIS	Geospatial/Geographic Information System
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
HCS	Host Computer System
IACO	International Civil Aviation Organization
IAF	Initial Approach Fix
IF	Intermediate Fix
IFR	Instrument Flight Rules
ILS	Instrument Landing System
IM	Inner Marker
IMC	Instrument Meteorological Conditions
LPV	Localizer Performance with Vertical Guidance
MHz	Megahertz
MOU	Memorandum of Understanding
MRO	Multiple Runway Operations
MRS	Minimum Required Separation
MSL	Mean-Sea Level

Copyright National Academy of Sciences. All rights reserved.

ACRONYM	DEFINITION
MTOW	Maximum Takeoff Weight
NAC	NextGen Advisory Committee
NAS	National Airspace System
NAVAID	Navigational Aid
NBAA	National Business Aviation Association
NDB	Non-directional Beacon
NEMS	NAS Enterprise Messaging System
NEPA	National Environmental Policy Act
NextGen	Next Generation Air Transportation System
NM	Nautical Mile
NOTAM	Notice to Airmen
NPIAS	National Plan of Integrated Airport Systems
NVS	NAS Voice System
OAPM	Optimization of Airspace and Procedures in the Metroplex
OPD	Optimized Profile Descents
PACI	Passenger Airline Cost Index
PBN	Performance-Based Navigation
PRM	Precision Runway Monitor
RAA	Regional Airline Association
RECAT	Recategorization of Separation Standards
RF	Radio Frequency
RNAV	Area Navigation
RNP	Required Navigation Performance
RTCA	Radio Technical Commission for Aeronautics
SID	Standard Instrument Departure
STAR	Standard Terminal Arrival Route
STARS	Standard Terminal Automation Replacement System
STDDS	SWIM Terminal Distribution System
SVS	Synthetic Vision System
SVT	Surface Visualization Tool
SWIM	System Wide Information Management
TAMR	Terminal Automation Modernization And Replacement
TBFM	Time-Based Flow Management
TBM	Time-Based Metering
TBO	Trajectory-Based Operations
TERPS	Terminal Procedures
TFDM	Terminal Flight Data Manager
TIS-B	Traffic Information System–Broadcast

Copyright National Academy of Sciences. All rights reserved.

ACRONYM	DEFINITION
TIS-R	Traffic Information System–Rebroadcast
TMA	Traffic Management Advisor
TMI	Traffic Management Initiatives
TRACON	Terminal Radar Approach Control
UAT	Universal Access Transceiver
VALE	Voluntary Airport Low Emissions
VDL	VHF Data Link
VFR	Visual Flight Rules
VHF	Very High Frequency
VOR	Very High Frequency Omni-Directional Range
WAAS	Wide-Area Augmentation System
Wake RECAT	Wake Vortex Recategorization
WAM	Wide-Area Multilateration
WTMA	Wake Turbulence Mitigation for Arrivals
WTMA-P	WTMA-Procedures
WTMD	Wake Turbulence Mitigation for Departures

Copyright National Academy of Sciences. All rights reserved.

B Glossary

TERM	DEFINITION
Above Ground Level	The altitude expressed in the actual number of feet measured with respect to the underlying ground surface. The distance of the aircraft above the ground.
Administrator for Airports	The Federal Aviation Administration's office responsible for reviewing and deciding on projects involving airports, overseeing their construction and operations ensuring compliance with federal regulations.
Air Traffic Control (ATC)	A service provided by ground-based controllers who direct aircraft on the ground and through controlled airspace, and that also can provide advisory services to aircraft in non-controlled airspace. The primary purpose of ATC is to promote the safe, orderly, and expeditious flow of air traffic.
Airport Cooperative Research Program (ACRP)	An applied research program that develops practical solutions to problems faced by airport operators. ACRP is managed by the Transportation Research Board (TRB) of the National Academies of Sciences, Engineering, and Medicine and sponsored by the FAA.
Area Navigation (RNAV)	A method of navigation that allows an aircraft to choose any course within a network of navigation beacons, rather than navigating directly to and from the beacons. This can conserve flight distance, reduce congestion, and allow flights into airports without beacons.
Authorization Required	Refers to required navigation performance (RNP) instrument approach procedures (IAP) with authorization required (AR). Authorization is typically associated with aircraft avionics equipment, operator requirements, and pilot training.
Automatic Dependent Surveillance Broadcast (ADS-B)	An element of the U.S. Next Generation Air Transportation System (NextGen), ADS-B is an air traffic surveillance technology that enables aircraft to be accurately tracked by air traffic controllers and other pilots without the need for conventional radar.
Aviation Environmental Design Tool (AEDT)	A comprehensive software system that dynamically models aircraft performance in space and time to produce estimates of noise, fuel burn, and emissions at global, regional, and local levels. AEDT is currently used by the U.S. government to consider the interdependencies between aviation-related noise, criteria/hazardous air pollutants/greenhouse gas emissions, and fuel consumption.
Clean Air Act (CAA)	A United States federal law, first enacted in 1955, with major revisions in 1970 and 1977, that is designed to protect human health and the environment from the effects of air pollution. Under the CAA, the Environmental Protection Agency (EPA) is required to establish national ambient air quality standards (NAAQS) to protect public health and public welfare and to regulate emissions of hazardous air pollutants. State and local governments monitor and enforce CAA regulations, with oversight by the EPA.

Copyright National Academy of Sciences. All rights reserved.

TERM	DEFINITION
Closely Spaced Parallel Runway Operations (CSPO)	A procedure used by air traffic controllers to space aircraft closer together on takeoff and landing at major U.S. airports for the purpose of increasing airspace capacity. Under CSPO, aircraft pairs arriving at an airport with parallel runways that are separated by 2,500 feet or less are staggered to observe 1.5 nm diagonal separations between leading and trailing aircraft on the separate runways.
Cockpit Display of Traffic Information (CDTI)	A CDTI is a generic display that provides the flight crew with surveillance information about other aircraft, including their position. Traffic information for a CDTI may be obtained from one or multiple sources, including ADS-B, traffic collision avoidance system (TCAS), and TIS-B. Direct air-to-air transmission of ADS-B messages supports display of proximate aircraft on a CDTI.
Code of Federal Regulations (CFR)	The codification of the general and permanent rules and regulations published in the Federal Register by the executive departments and agencies of the federal government of the United States.
Day-Night Average Sound Level (DNL)	Expressed in decibels (dB), DNL is a 24-hour average noise level used to define the level of noise exposure on a community. The DNL represents the average sound exposure during a 24-hour period and does not represent the sound level for a specific noise event. A 10 dB correction is applied to nighttime (10:00 p.m. and 7:00 a.m.) sound levels to account for increased annoyance due to noise during the night hours.
Decibels (dB)	The logarithmic unit used to measure the intensity of a sound measuring from the threshold of human hearing, 0 dB, upward toward the threshold of pain, about 120–140 dB. An increase of 10 dB is perceived by human ears as a doubling of noise.
Distance Measuring Equipment (DME)	Equipment (ground and airborne) used to measure and report to the pilot the slant range distance, in nm, of an aircraft from the DME navigational aid.
Environmental Assessment (EA)	Assessment performed under U.S. environmental law, the National Environmental Policy Act (NEPA), used to predict the environmental consequences of a plan, policy, program, or project prior to the decision to move forward with the proposed action. The EA will determine either the need to prepare an environmental impact statement (EIS) or justify a finding of no significant impact (FONSI).
Environmental Impact Statement (EIS)	A document required by NEPA for certain actions that may significantly affect the quality of the human environment. The purpose of an EIS is to analyze and disclose the significant effects resulting from a Federal action, and also list alternative actions that may be chosen instead of the action described in the EIS.
Equivalent Lateral Spacing Operations (ELSO)	A procedure used by air traffic controllers to space aircraft closer together on takeoff and landing at major U.S. airports for the purpose of increasing airspace capacity. ELSO reduces the divergence angle between the departure routes of aircraft on takeoff, therefore allowing controllers to space routes more closely together and clear aircraft for takeoff more efficiently.
Flight-Deck-Based Interval Management-Spacing (FIM-S)	A component of interval management (IM), which is a set of applications that enable more precise and consistent spacing between aircraft to yield increased throughput and efficiency. The IM system is composed of a ground-based component (GIM) and a flight-deck-based component (FIM). The FIM component involves the use of avionics, which provides guidance to achieve and/or maintain a desired spacing interval relative to a target aircraft.
Flight Information Services–Broadcast	A component of ADS-B technology that provides free graphical National Weather Service products, temporary flight restrictions (TFRs), and special-use airspace information.

Copyright National Academy of Sciences. All rights reserved.

TERM	DEFINITION
Flight Management System	A suite of avionics programs on board an aircraft used to calculate the most economical flying speeds and altitudes during a flight and to identify possible choices in emergencies.
General Aviation	All civil aviation operations other than scheduled air services and non-scheduled air transport operations for remuneration or hire. The majority of the world's air traffic falls into this category, and most of the world's airports serve GA exclusively.
Global Positioning System (GPS)	A system of satellites, computers, and receivers that is able to determine the latitude and longitude of a receiver on Earth by calculating the time difference for signals from different satellites to reach the receiver. In aviation, GPS data allows pilots to obtain precise 3D or 4D location data.
Government Accountability Office (GAO)	An independent, nonpartisan agency established by the Budget and Accounting Act of 1921 that investigates how the federal government spends taxpayer dollars.
Ground-Based Augmentation System (GBAS)	A system that provides differential corrections and integrity monitoring of global navigation satellite systems. GBAS provides navigation and precision approach service in the vicinity of the host airport, broadcasting its differential correction message via a very high-frequency radio data link from a ground-based transmitter. GBAS yields the extremely high accuracy, availability, and integrity necessary for Category I, and eventually Category II and III precision approaches.
Inertial Reference Unit (IRU)	A type of inertial sensor which uses only gyroscopes to determine a moving aircraft's change in angular direction over a period of time.
Instrument Flight Procedure (IFP)	A description of a series of predetermined flight maneuvers by reference to flight instruments, published by electronic and/or printed means.
Instrument Landing System (ILS)	A radar-based instrument approach system that provides precision lateral and vertical guidance to ILS-equipped aircraft approaching and landing on a runway, enabling a safe landing during instrument meteorological conditions (IMC), such as low ceilings or reduced visibility.
Integrated Noise Model (INM)	The Integrated Noise Model (INM) is a computer model that evaluates aircraft noise impacts in the vicinity of airports. The INM can output either noise contours for an area or noise level at pre-selected locations. In the U.S., INM is the preferred model used for FAR Part 150 noise compatibility planning and for FAA Order 1050 ESs and EISs.
In-trail Procedures (ITP)	An ADS-B application developed by the FAA. The use of flight-level change procedures, enabled by ADS-B ITP, enables flight-level changes for aircraft operating in oceanic airspace and being held at non-optimal flight levels due to conflicting traffic.
Lateral Navigation (LNAV)	GPS-based non-precision instrument approach procedure that provides horizontal approach navigation without approved vertical guidance. The approach minimums for LNAV approaches are higher than that of ILS approaches and RNAV approaches that incorporate vertical guidance.
Localizer Performance (LP)	An RNAV function using a final approach segment data block that computes, displays, and provides horizontal approach navigation using the horizontal accuracy and integrity of localizer performance with vertical guidance (LPV) without approved vertical guidance. The LP line of minima is provided at locations where issues prevent the use of vertical guidance, and provides a higher probability of achieving the lowest minimum at these locations.
Localizer Performance with Vertical Guidance (LPV)	An RNAV function using a final approach segment data block, which computes, displays, and provides both horizontal and approved vertical approach navigation to minimums as low as 200 foot ceiling and ½-mile visibility.

Copyright National Academy of Sciences. All rights reserved.

TERM	DEFINITION
Metroplex	Metroplex refers to a system of airports in close proximity and their shared airspace that serve one or more major cities.
National Airspace System (NAS)	Created by FAA, to protect persons and property on the ground, and to establish a safe and efficient airspace environment for civil, commercial, and military aviation. The NAS is made up of a network of air navigation facilities, ATC facilities, airports, technology, and appropriate rules and regulations that are needed to operate the system.
National Environmental Protection Act (NEPA)	A congressional Act that established the national policy for disclosing the potential impacts of federal actions. Compliance with NEPA requires the completion of an environmental document that outlines impacts that may significantly affect the quality of the human environment.
Optimal Profile Descent (OPD)	An aircraft approach method designed to reduce fuel consumption and noise compared to other conventional descents. Instead of approaching an airport in a stair-step fashion, OPD allows for a smooth, constant-angle descent to landing.
Performance-Based Navigation (PBN)	A term used to describe the broad range of technologies that move aviation away from a ground-based navigation system toward a system that relies more on the performance and capabilities of equipment on board the aircraft. PBN specifies that aircraft RNP and RNAV systems performance requirements be defined in terms of accuracy, integrity, availability, continuity, and functionality required for the proposed operations.
Required Navigation Performance (RNP)	A type of PBN that allows an aircraft to fly a specific path between two 3D-defined points in space. RNP equipment provides onboard navigation capability that allows crews to accurately fly aircraft along a precise flight path. RNP also refers to the level of performance required for a specific procedure or a specific block of airspace. An RNP of 10 means that a navigation system must be able to calculate its position to within a circle with a radius of 10 nm.
RTCA	A nonprofit U.S. volunteer organization that develops technical guidance for use by government regulatory authorities and by industry. Originally named the Radio Technical Commission for Aeronautics, it is now officially known solely as the RTCA.
Standard Instrument Approach Procedures (SIAP)	A series of predetermined maneuvers for the orderly transfer of an aircraft under instrument flight conditions from the beginning of the initial approach to a landing or to a point from which a landing may be made visually or the missed approach procedure is initiated. It is prescribed and approved for a specific airport by a competent authority.
Standard Instrument Departures	A preplanned instrument flight rule (IFR) departure procedure published for pilot use, in graphic or textual format, that provides obstruction clearance from the terminal area to the appropriate en route structure.
Standard Terminal Arrival (STA)	A published IFR arrival procedure describing specific criteria for descent, routing, and communications for a specific runway at an airport.
Terminal Radar Approach Control (TRACON)	An FAA ATC facility that uses radar and two-way radio communication to provide separation of air traffic within a specified geographic area in the vicinity of one or more large airports.
Traffic Collision Avoidance System (TCAS)	An aircraft collision avoidance system designed to reduce the incidence of mid-air collisions between aircraft. TCAS is based on secondary surveillance radar transponder signals, and operates independently of ground-based equipment to provide advice to the pilot on potential conflicting aircraft. The ICAO mandates that the system be fitted to all aircraft with a maximum take-off mass of over 5,700 kg (12,600 lb) or authorized to carry more than 19 passengers.

Copyright National Academy of Sciences. All rights reserved.

TERM	DEFINITION
Traffic Information Services-Broadcast (TIS-B)	TIS-B is the broadcast of ATC-derived traffic information to ADS-B-equipped (1090ES or UAT) aircraft from ground radio stations. The source of this traffic information is derived from ground-based air traffic surveillance radar sensors. TIS-B service will be available throughout the NAS where there are both adequate surveillance coverage (radar) from ground sensors and adequate broadcast coverage from ADS-B ground radio stations.
Vertical Navigation (VNAV)	A form of precise vertical (altitude) navigation using the aircraft flight management system (FMS). VNAV is the vertical navigation flight profile, which is the predicted flight trajectory of the airplane in the vertical plane as a function of distance along the horizontal flight path defined by the LNAV flight plan.
Wide-Area Augmentation System (WAAS)	An air navigation aid developed by the Federal Aviation Administration to augment GPS, with the goal of improving its accuracy, integrity, and availability. WAAS is intended to enable aircraft to rely on GPS for all phases of flight, including precision approaches.
Wide-Area Multilateration (WAM)	A NextGen surveillance capability that enables air traffic controllers to track aircraft flying into and out of airports in mountainous areas with no radar coverage. WAM can complement ADS-B by providing transitional surveillance for non-ADS-B-equipped targets, and can be used for ADS-B validation.

Copyright National Academy of Sciences. All rights reserved.

C Public Information Toolkit

Introduction

The Public Information Toolkit provides information on what and how to communicate with a broad spectrum of external stakeholders, such as local politicians, business leaders, the surrounding community, local environmental advocates, the local pilot community, and others with aviation interests. It provides a number of direct and simple messages about NextGen and expected changes from NextGen. Note that the full toolkit is available on the ACRP NextGen series site: https://crp.trb.org/acrp0127.

NextGen Messages

For these audiences, there are two sets of messages that overlap across groups. The first is that air travel can improve and grow as a result of NextGen, with business impacts and greater reliability of air travel. The following statements can be made about NextGen to an airport's stakeholders:

- NextGen will enable more efficient flight routes with the potential for lower fuel burn.
- These efficiency improvements represent billions of dollars of saved fuel and time to airlines, but in airport terms, traffic will increase only in relatively small amounts year by year.
- NextGen enables more efficient use of airspace, which can increase throughput at an airport even if the airport does nothing. Actual increases in airport traffic will depend, as before, on airline decisions, local economy impacts, and the overall economy.
- A NextGen Metroplex program includes the air traffic procedures at nearby GA, reliever, and small airports, so that flights to and from smaller nearby airports will be easier for aircraft with GPS.
- Flights to mountainous airports and foggy areas will become more reliable.
- Flights in bad weather will be delayed less and cancelled less.

The second set of messages is the issues of interest in the community surrounding the airport, and issues for those with environmental interests. NextGen has the potential to bring changes to these communities that can be seen as either good or bad. Broadly, public information concerns will fall into three categories:

- The pattern of aircraft noise may change. Those who already experience aircraft noise may see it increase or decrease.
- The period of environmental review for the new procedures and flight pattern changes may appear to be shortened.
- Greenhouse gas emissions on airports are being reduced.

Copyright National Academy of Sciences. All rights reserved.

Communicating with Community

As noted in this resource guide and *ACRP Report 150: NextGen for Airports: A Primer*, FAA notification procedures for changed routes that alter noise patterns may occur far enough in advance that the community forgets about them, or is unaware of them. Airports can promote good neighbor relations by providing consistent and ongoing communication about upcoming changes, raising awareness of both the positive environmental and access benefits of NextGen as well as the potential for changed noise patterns. The materials in this Toolkit will provide messaging examples that can be employed with the community. See *ACRP Report 150: NextGen for Airports, Volume 2: Engaging Airport Stakeholders: Guidebook* for additional suggestions on outreach strategies.

The toolkit provides the following information:

- Single page flyers and short brochures that can be used by airport in their outreach (the flyers are reprinted on the following pages for your convenience).
- An animated presentation accessible electronically.

Copyright National Academy of Sciences. All rights reserved.

Copyright National Academy of Sciences. All rights reserved.

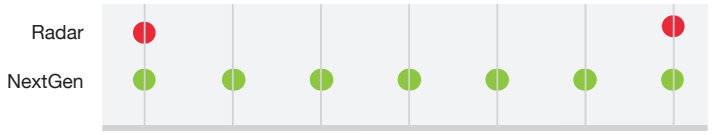
NextGen: An Overview

NextGen, the Next Generation Air Transportation System, is a set of programs being implemented by the Federal Aviation Administration (FAA) over the next several years. This modernization program is a complex undertaking that the FAA believes will lead to more efficient and accurate flight trajectories that will benefit travelers and airlines.

SOME BACKGROUND

Since the 1950s, airplanes have been guided by radar systems. While there have been steady improvements to the technology, satellite-based Global Positioning Systems (GPSs) can deliver more precise position information. For example, due to radar tracking's limitations, planes maintain a 5-mile buffer in the air and a 3-mile one on landings.

NextGen Updates Positions Much More frequently



Radar-based systems update positions once every 6–12 seconds. However, NextGen-based systems update every second, making flying safer.

GPS and digital technologies will change that. They provide the precise position of an airplane, whether over land or water, to air traffic controllers and other aircraft within 150 miles. GPS also updates the aircraft position every second, which is more frequently than the 6 to 12 seconds with radar.

ADVANCING AIR SAFETY

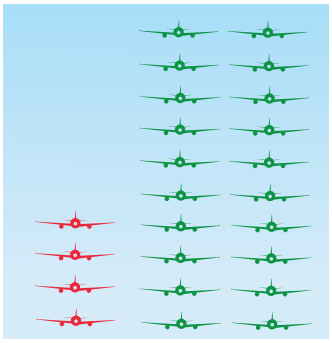
The new navigation procedures enabled by NextGen will improve the management of air traffic, making our already safe skies even safer. NextGen also enables shorter and more direct flight paths. These, too, increase system safety.

BOOSTING EFFICIENCY

Weather has often been a major factor in flight delays. With GPSs, airports can now support take-offs and landings when low clouds would have limited or prevented aircraft from flying or landing.

NextGen affects every aspect of the air travel experience, from departure gate to arrival gate. Knowing the exact position of aircraft on the ground allows better timing of gate pushback, reduces stacked airplanes waiting

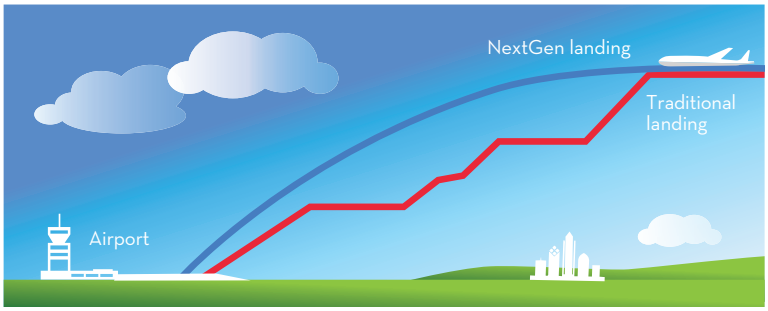
Slope Time



Ski season in Colorado brings big business from tourism. Traditionally, unpredictable weather from November to April meant about 75 flight delays per day at some close-to-the-slopes airports. Complaints were rampant. Frustrations were elevated. Following the installation of critical NextGen infrastructure the complaints have dropped to zero. How precise is it? Before NextGen controllers could only accept four arrivals and departures per hour. With NextGen: 15 to 20 per hour.

Copyright National Academy of Sciences. All rights reserved.

for takeoff, and enables controllers to more efficiently guide aircraft to the gate.

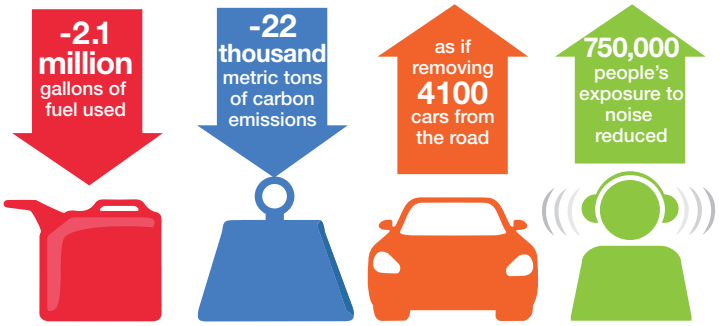


The Greener Skies Over Seattle initiative has demonstrated environmental benefits.

The new system allows pilots to land in a continuous descent approach, unlike the traditional "stairstep" approach currently used in some locations. This new descent is much smoother as airplanes are better able to glide in at reduced thrust.

IMPROVING THE ENVIRONMENT

With NextGen, airport takeoffs and landings follow more precise flight tracks. Continuous descent reduces nitrous oxide emissions. Along with better management of airplanes on the ground, this means less fuel burned and lower exposure to noise from gate to gate.



FIND OUT MORE

Visit www.faa.gov/nextgen/ to learn more.

AIRPORT LOGO
CAN BE PLACED HERE

KNOWING nextGen

Copyright National Academy of Sciences. All rights reserved.

NextGen Is Changing Air Travel

NextGen, the Next Generation Air Transportation System, is a set of programs being implemented by the Federal Aviation Administration (FAA) over the next several years. This modernization program is a complex undertaking that the FAA believes will lead to more efficient and accurate flight trajectories that will benefit travelers and airlines.

NextGen replaces ground-based tracking systems with satellite-based Global Positioning Systems (GPSs). This change allows more precise, accurate routing and tracking of aircraft, on the ground and in the air. That can mean that existing airport infrastructure can support more airport activity. However, this will not happen immediately. It may be several years before communities experience increased airport activity.

AIRPORTS PROVIDE ECONOMIC BENEFIT

Airports are a significant contributor to local economies. In 2013 the 485 commercial airports across the U.S. supported 9.6 million jobs totaling $358 billion in payroll.[1] From grounds crews to concessionaires, airport activity provides jobs for your neighbors and friends.

NEXTGEN SUPPORTS ENVIRONMENTALLY HEALTHY GROWTH

One major concern with airport activity is its environmental impact. With NextGen, airport takeoffs and landings follow more precise flight tracks. Continuous descent reduces nitrous oxide emissions. Along with better management of airplanes on the ground, this means less fuel burned and lower exposure to noise from gate to gate.

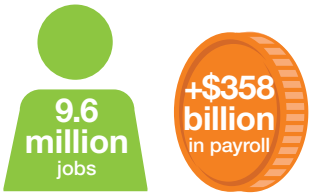
A WORK IN PROGRESS

In order to make the entire system more environmentally positive, some individuals and communities will notice increased noise as air traffic is concentrated along a narrower path. Our airport will work closely with the FAA as they develop NextGen procedures for our airport to maximize the environmental benefit and minimize negative impacts.

FIND OUT MORE

Visit www.faa.gov/nextgen/ to learn more.

The Contribution of All 485 Commercial Airports in the U.S.



Potential Environmental Benefits



AIRPORT LOGO
CAN BE PLACED HERE

KNOWING nextGen

[1] "The Economic Impact of Commercial Airports in 2013," Prepared for Airports Council International—North America by CDM Smith, September 2014.

Copyright National Academy of Sciences. All rights reserved.

Copyright National Academy of Sciences. All rights reserved.

NextGen Is Changing Air Travel

NextGen, the Next Generation Air Transportation System, is a set of programs being implemented by the Federal Aviation Administration (FAA) over the next several years. This modernization program is a complex undertaking that the FAA believes will lead to more efficient and accurate flight trajectories that will benefit travelers and airlines.

NextGen replaces ground-based tracking systems with satellite-based Global Positioning Systems (GPSs). This change allows more precise, accurate routing and tracking of aircraft, on the ground and in the air.

AIRPORT OPERATIONS WILL IMPROVE

Implementing NextGen will bring three major changes to airport operations:

- New technologies will change flight paths, enabling aircraft to fly more direct routes, saving fuel and reducing CO_2 emissions.
- Tracking technologies will pinpoint aircraft in the air and on the ground, allowing for better aircraft management.
- Data sharing will provide airports with a better picture of aircraft operations and enable better system-wide planning, especially in inclement weather.

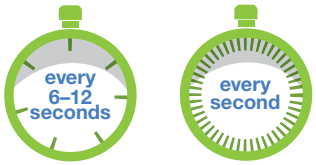
NEXTGEN PINPOINTS AIRCRAFT LOCATIONS



GPS technology allows for more precision.

Just one change – reducing air travel delays caused by weather – will have a significant impact on the air travel system. Delays at large airports cause ripple effects throughout the system, impacting airports of every size. NextGen technology addresses this by pinpointing aircraft position relative to weather systems and precisely rerouting travel to avoid weather problems.

AIRCRAFT POSITIONING UPDATE FREQUENCY



Radar-based | NextGen-based

70%

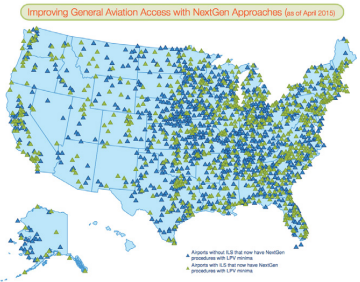
PERCENTAGE OF DELAYS THAT ARE ATTRIBUTABLE TO WEATHER

Weather has often been a major factor in flight delays. According to the FAA, 70% of flight delays can be attributed to weather. In terminal areas, NextGen technologies allow continued operation during inclement weather. With NextGen, it is possible to have safe arrival and departure windows that would previously have been closed due to weather.

Copyright National Academy of Sciences. All rights reserved.

SMALLER AIRPORTS WILL ALSO BENEFIT

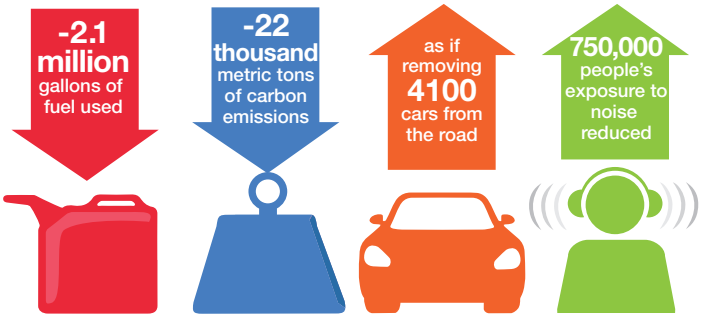
NextGen improvements will be seen in smaller and general aviation airports, too. The same technology enhancements happening at large metroplex airports will mean better instrument approaches and safer air travel. General aviation pilots who install an Automatic Dependent Surveillance–Broadcast (ADS-B) unit in their aircraft will receive weather and aircraft traffic information right in their cockpit. For example, it's already helping ADS-B-equipped pilots at Charles M. Schulz in Sonoma County, California, land their aircraft when the approach is covered in fog.



NEXTGEN SUPPORTS ENVIRONMENTALLY HEALTHY GROWTH

With NextGen, airport takeoffs and landings follow more precise flight tracks. Continuous descent reduces nitrous oxide emissions. Along with better management of taxi queues on the ground, this means less fuel burned from gate to gate.

IN SEATTLE, NEXTGEN IMPROVED THE ENVIRONMENT



NEXTGEN ALSO RAISES SOME CONCERNS

Any change this large and complex will affect our community. Our airport will work closely with the FAA as they develop NextGen procedures for our airport to maximize the environmental benefit and minimize impacts.

FIND OUT MORE

Visit www.faa.gov/nextgen/ to learn more.

AIRPORT LOGO
CAN BE PLACED HERE

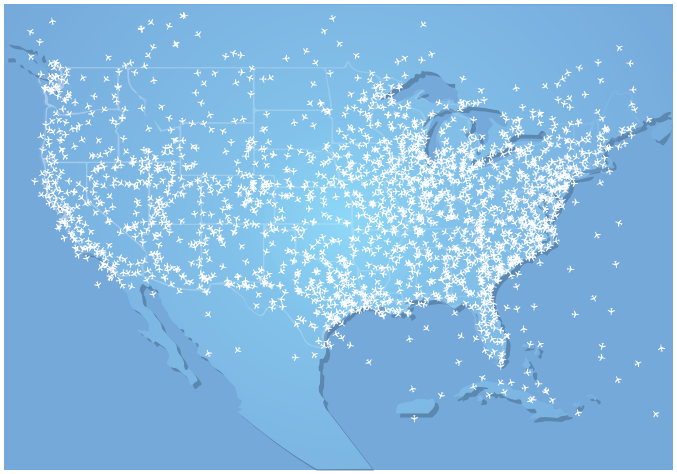
KNOWING nextGen

Copyright National Academy of Sciences. All rights reserved.

NextGen: Advancing Air Safety

At any given time, there are around 7,000 aircraft in the air in the U.S. To monitor all of that traffic and keep it moving safely, air traffic controllers have relied on monitoring screens showing information gathered from ground-based tracking and radar systems, a process that has been used since the 1950s. While it is quite reliable, advances in technology, particularly satellite-based Global Positioning Systems (GPSs), are enabling a new era in managing the air transportation system. Known as NextGen, the Next Generation Air Transportation System, it is being implemented by the Federal Aviation Administration (FAA).

NextGen Increases Safety Across the Nation



More than 1,600 airports in the U.S. already have NextGen procedures making it easier for all pilots, including general aviation pilots, to fly to airports during low visibility conditions.

PRECISE INFORMATION IMPROVES SAFETY

Pilots can use flight deck displays of other aircraft positions to maintain proper separation, such as when landing on closely spaced runways. This information is available under Automatic Dependent Surveillance–Broadcast (ADS-B), which reports aircraft location derived from GPS. This can also permit taxi procedures in low visibility, reducing weather-related delays, and maintaining more predictable operations.

NextGen also enables more direct, shorter flight paths. Improved GPS-based surveillance plus predictable, repeatable flight paths will allow

70%

PERCENTAGE OF DELAYS THAT ARE ATTRIBUTABLE TO WEATHER

Weather has often been a major factor in flight delays. According to the FAA, 70% of flight delays can be attributed to weather. In terminal areas, NextGen technologies allow continued operation during inclement weather. With NextGen, it is possible to have safe arrival and departure windows that would previously have been closed due to weather.

Copyright National Academy of Sciences. All rights reserved.

the air travel system to continue current levels of safety in the face of increased future traffic demand.

Weather has often been a major factor in flight delays. In fact, according to the FAA, 70% of flight delays can be attributed to weather. Now, with NextGen, airplanes can share exact weather data with controllers and reroute around weather systems. In terminal areas, NextGen technologies allow continued operations during inclement weather. With NextGen, it is possible to have safe arrival and departure windows that would previously have been closed due to weather.

NextGen Pinpoints Aircraft Locations

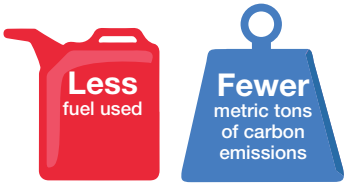


GPS technology allows for more precision.

FIND OUT MORE

Visit www.faa.gov/nextgen/ to learn more.

In Washington, D.C., NextGen Saved Fuel and Reduced Pollution



AIRPORT LOGO
CAN BE PLACED HERE

KNOWING
nextGen

Copyright National Academy of Sciences. All rights reserved.

NextGen: Better for the Environment

NextGen, the Next Generation Air Transportation System, is a set of programs being implemented by the Federal Aviation Administration (FAA) over the next several years. It includes protecting the environment while allowing for growth in air travel.



EFFICIENCY IMPROVES AIR QUALITY

NextGen replaces ground-based radar data with satellite-based data from the U.S. Global Positioning Satellite System, or GPS. Because it is more precise than the systems that have been used for years, it improves the efficiency of the country's air transportation system, using less fuel and reducing emissions.

From gate to gate, fuel improvements are noticeable. Ground operations can be managed to reduce waiting on the tarmac. In flight, the routes enabled by GPS can mean shorter, more direct flights that use less fuel.

According to the FAA, "[a]irlines using NextGen procedures at Washington, D.C., airports are expected to save 2.3 million gallons of fuel per year and cut aircraft exhaust emissions by 7,300 metric tons." (Source: FAA Nextgen Update: 2014)

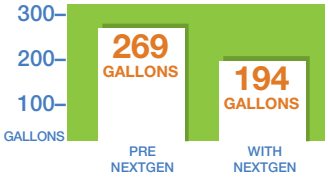
REDUCING NOISE

More precision brings tighter takeoff and landing patterns, which means a smaller area is impacted by airport noise. In Seattle, for example, the FAA, airlines, the Port of Seattle, and Boeing Corporation are collaborating on a project that has added 27 new procedures to reduce noise and emissions. Alaska Airlines estimates that the Greener Skies Over Seattle initiative, as the project is known, will cut fuel consumption by 2.1 million gallons annually and reduce carbon emissions by 22,000 metric tons. That's like taking 4,100 cars off the road every year. And it will reduce noise exposure for about 750,000 people living along the flight corridor.

"…airlines using NextGen procedures at Washington, D.C., airports are expected to save 2.3 million gallons of fuel per year and cut aircraft exhaust emissions by 7,300 metric tons."

— FAA Nextgen Update: 2014

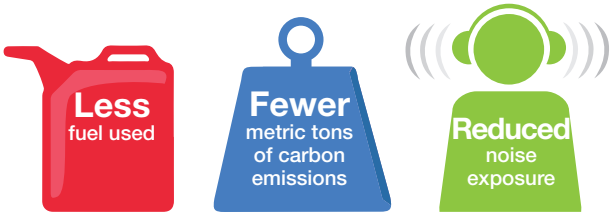
Example of NextGen Fuel Saving on Landing



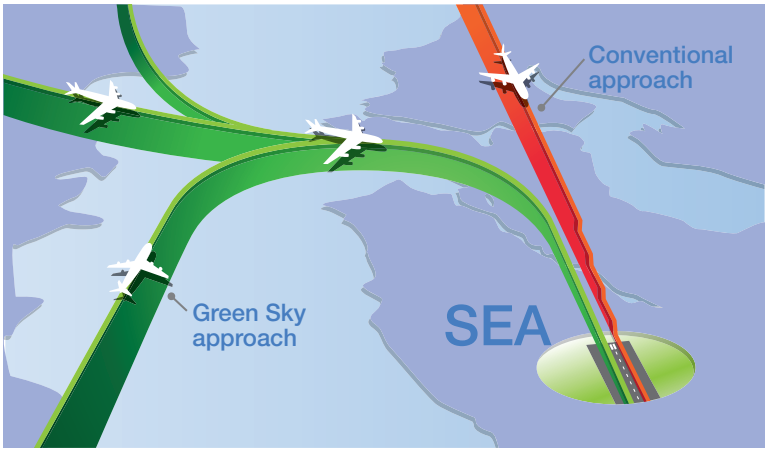
A flight from Phoenix Sky Harbor International Airport to Ronald Reagan Washington National Airport using NextGen technology demonstrates how fuel is saved. As US Airways Captain Brian Townsend pulled back the throttles to begin his descent from 37,000 feet, he had 1,493 gallons of fuel onboard. When he landed the Airbus A320 27 minutes later, he had only burned 194 gallons, less than the 269 gallons that had been the average.

Copyright National Academy of Sciences. All rights reserved.

In Seattle, NextGen Improved the Environment



Green Skies Over Seattle Initiative Flight Paths



The system isn't perfect. In order to make the entire system more environmentally positive, some individuals and communities will notice increased noise as air traffic is concentrated along a narrower path. Our airport will work closely with the FAA as they develop NextGen procedures for our airport to maximize the environmental benefit and avoid negative impacts.

FIND OUT MORE

Visit www.faa.gov/nextgen/ to learn more.

AIRPORT LOGO
CAN BE PLACED HERE

KNOWING
nextGen

Copyright National Academy of Sciences. All rights reserved.

Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S.DOT	United States Department of Transportation

Copyright National Academy of Sciences. All rights reserved.